

Results Report Homework 1

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1 Default Scenario

Scenario type: default reference run. This baseline scenario uses four training procedures with heterogeneous worker counts, demand values, periods, and phase offsets. The result folders contain the simple, normal, advanced, and temporal strategies.

Training	Workers	Collector	D	T	ϕ	Cycles
Blue	10	c1 (10.0.1.1)	50	30	1	4
Green	8	c2 (10.0.1.2)	62.5	40	2.5	4
Red	6	c3 (10.0.1.3)	83.35	30	4	4
Yellow	4	c4 (10.0.1.4)	125	40	5	4

1.1 Simple Strategy

The following plots report all available outputs for the `simple` strategy in `default_scenario`.

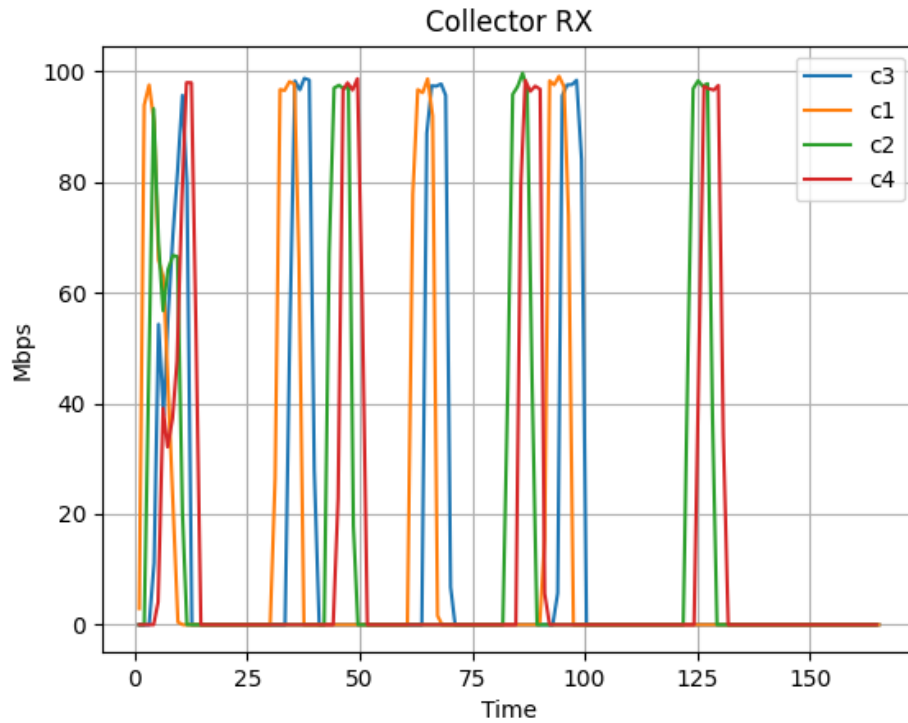


Figure 1: Default Scenario – Simple strategy, Global result.

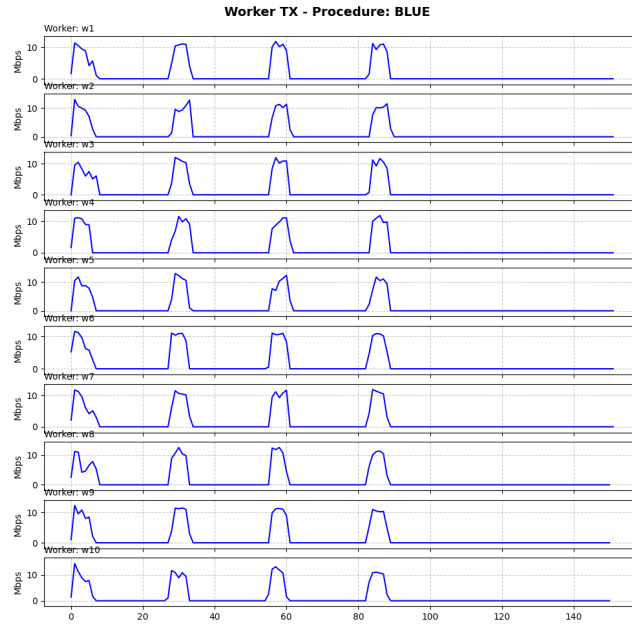


Figure 2: Default Scenario – Simple strategy, Blue result.

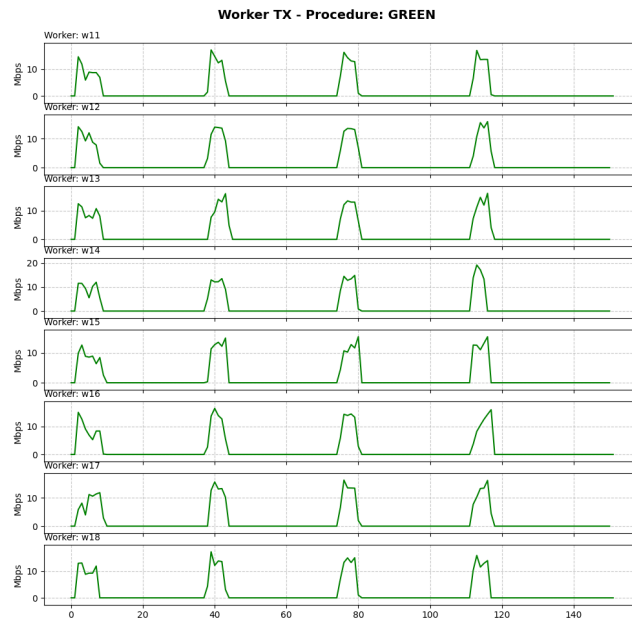


Figure 3: Default Scenario – Simple strategy, Green result.

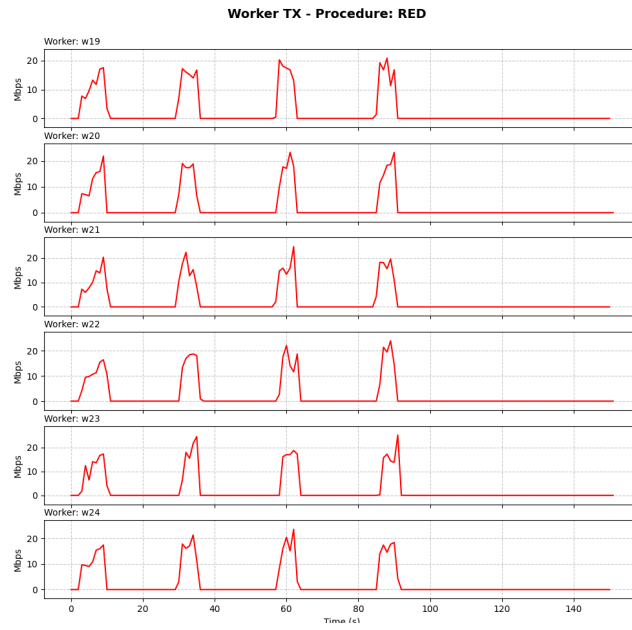


Figure 4: Default Scenario – Simple strategy, Red result.

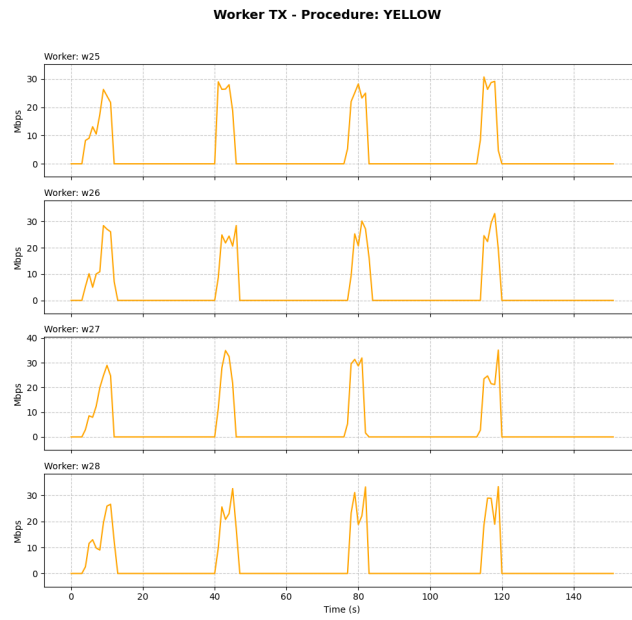


Figure 5: Default Scenario – Simple strategy, Yellow result.

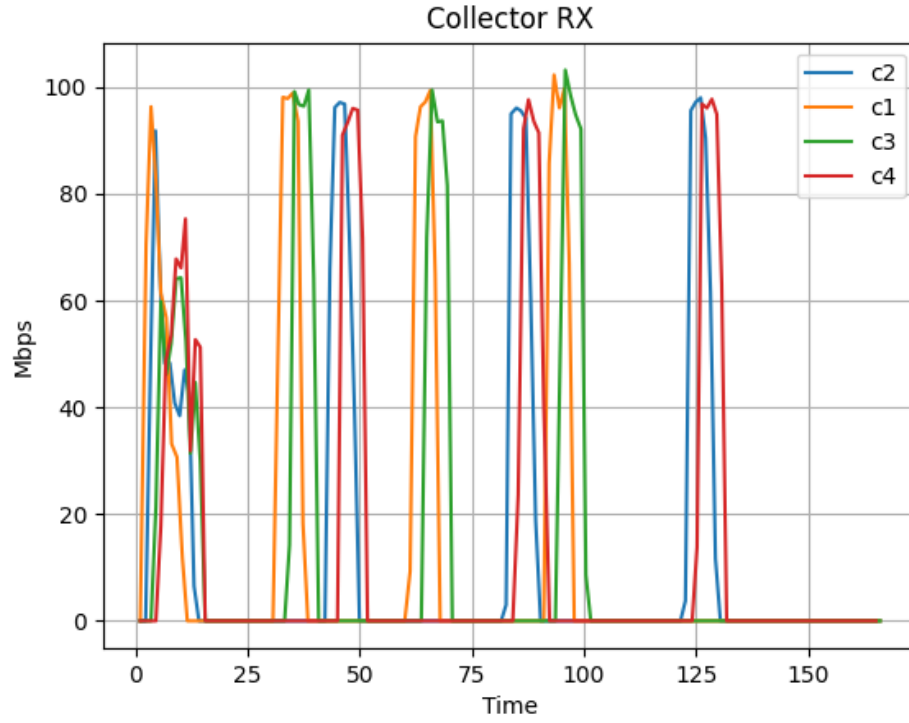


Figure 6: Default Scenario – Simple strategy, Global2 result.

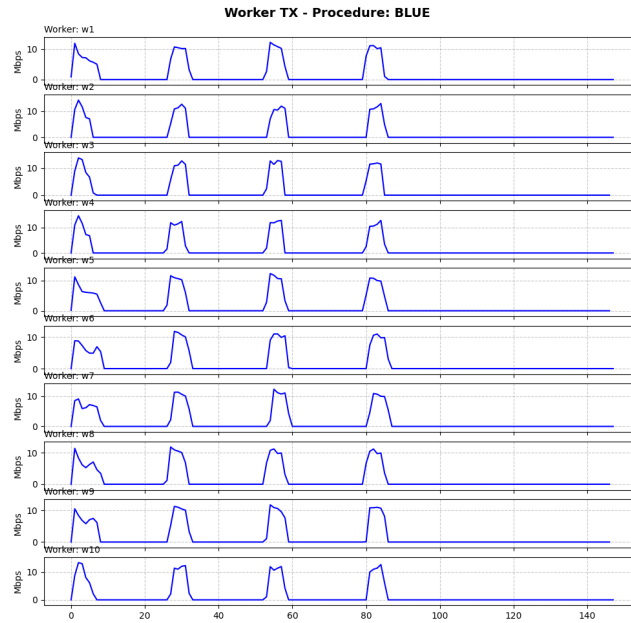


Figure 7: Default Scenario – Simple strategy, Blue2 result.

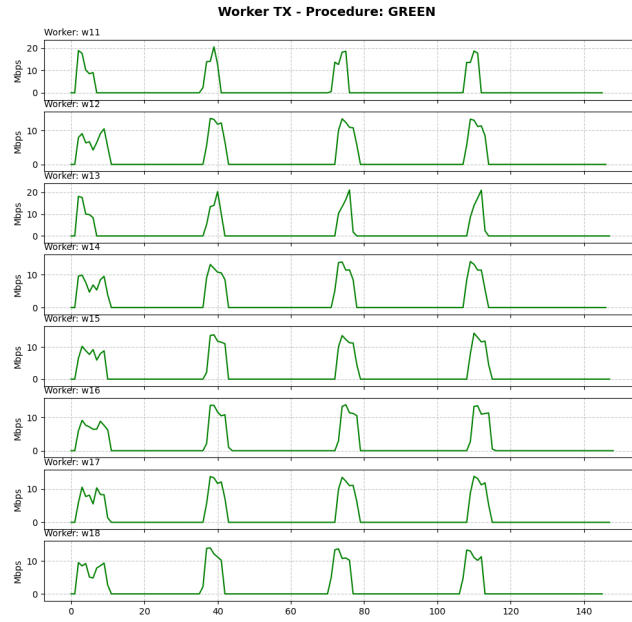


Figure 8: Default Scenario – Simple strategy, Green2 result.

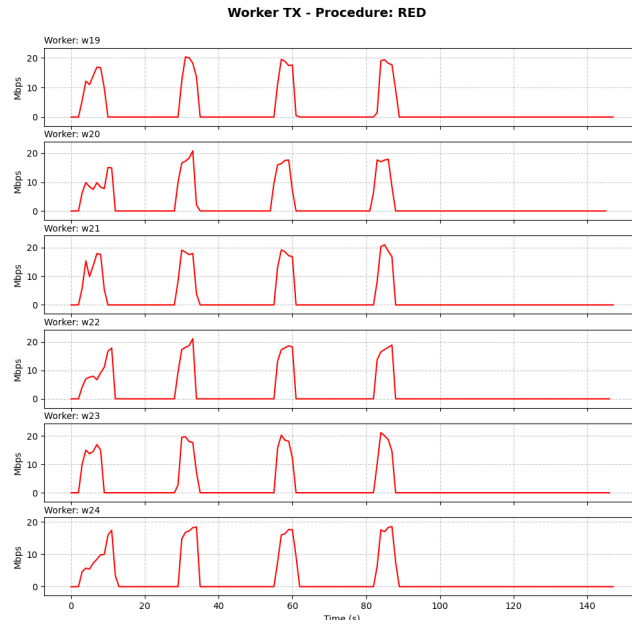


Figure 9: Default Scenario – Simple strategy, Red2 result.

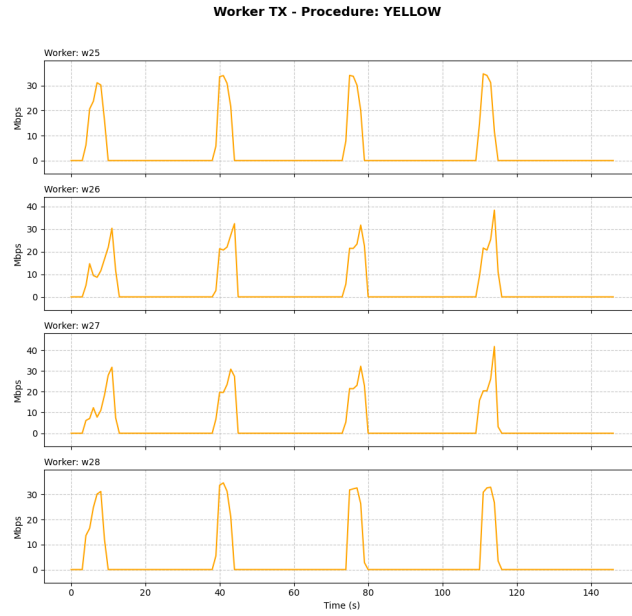


Figure 10: Default Scenario – Simple strategy, Yellow2 result.

1.2 Normal Strategy

The following plots report all available outputs for the `normal` strategy in `default_scenario`.

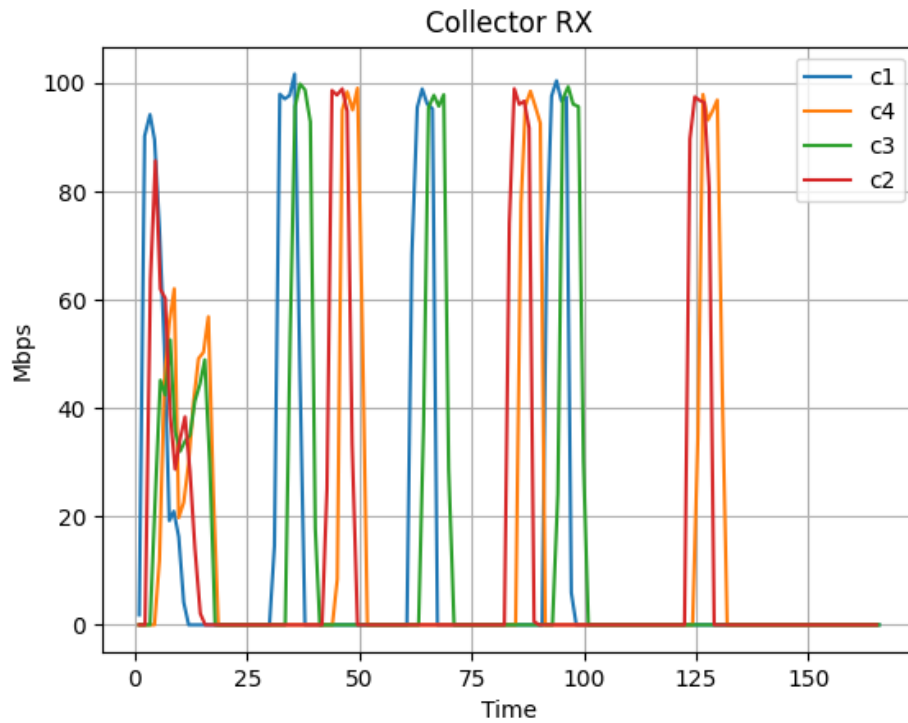


Figure 11: Default Scenario – Normal strategy, Global result.

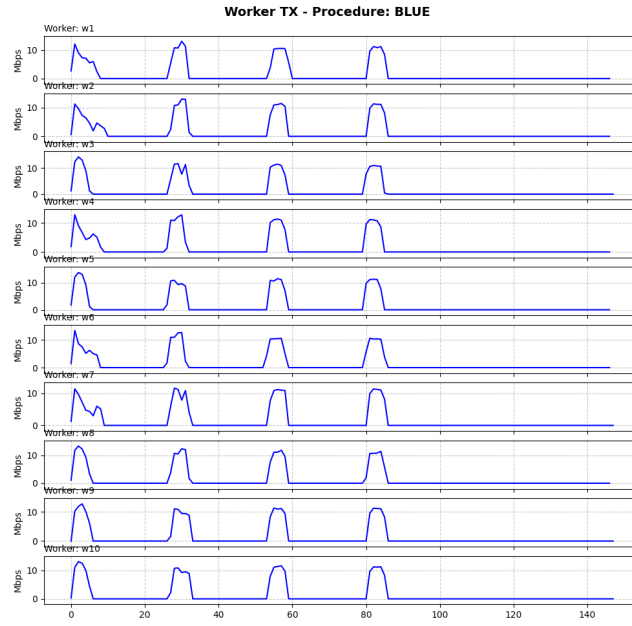


Figure 12: Default Scenario – Normal strategy, Blue result.

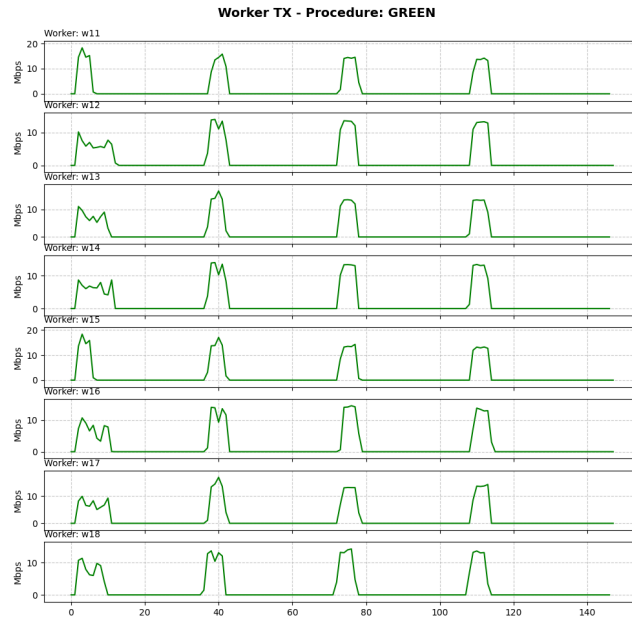


Figure 13: Default Scenario – Normal strategy, Green result.

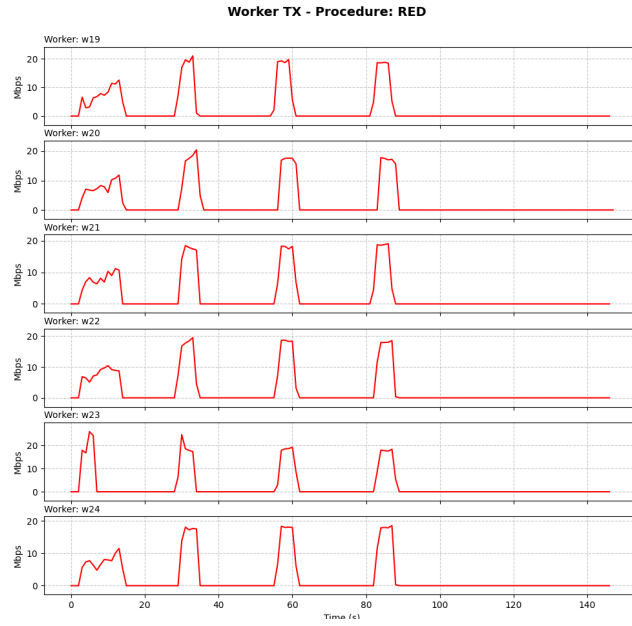


Figure 14: Default Scenario – Normal strategy, Red result.

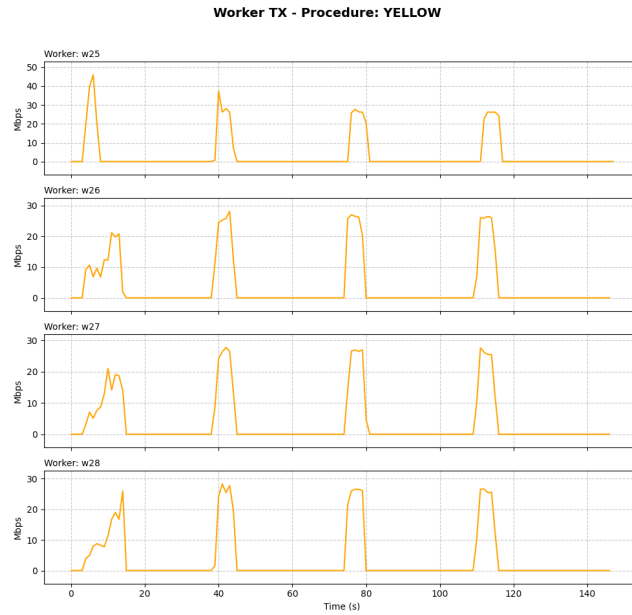


Figure 15: Default Scenario – Normal strategy, Yellow result.

1.3 Advanced Strategy

The following plots report all available outputs for the `advanced` strategy in `default_scenario`.

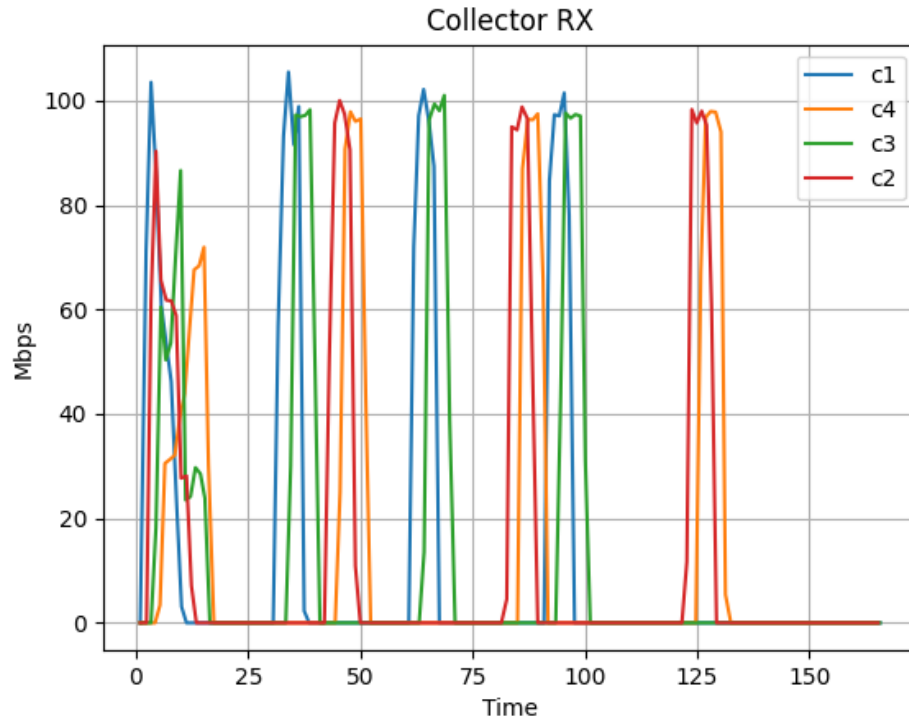


Figure 16: Default Scenario – Advanced strategy, Global result.

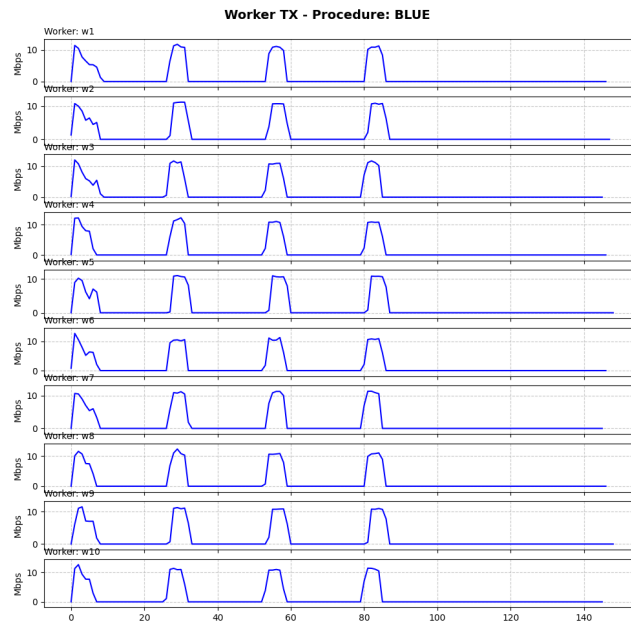


Figure 17: Default Scenario – Advanced strategy, Blue result.

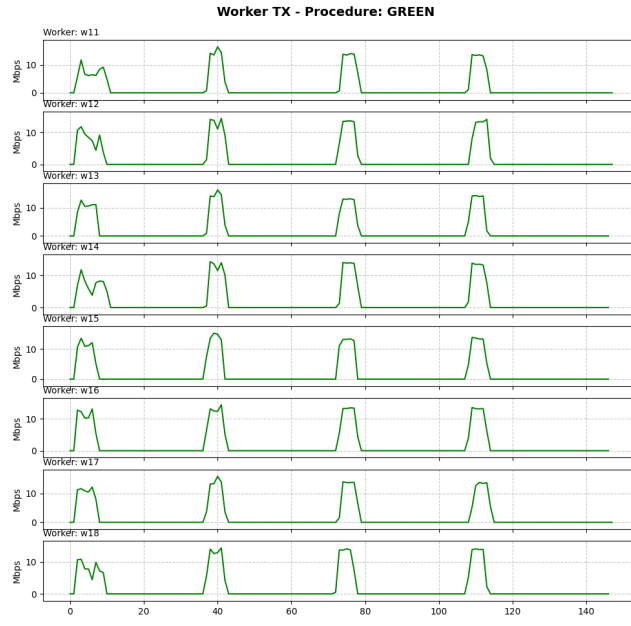


Figure 18: Default Scenario – Advanced strategy, Green result.

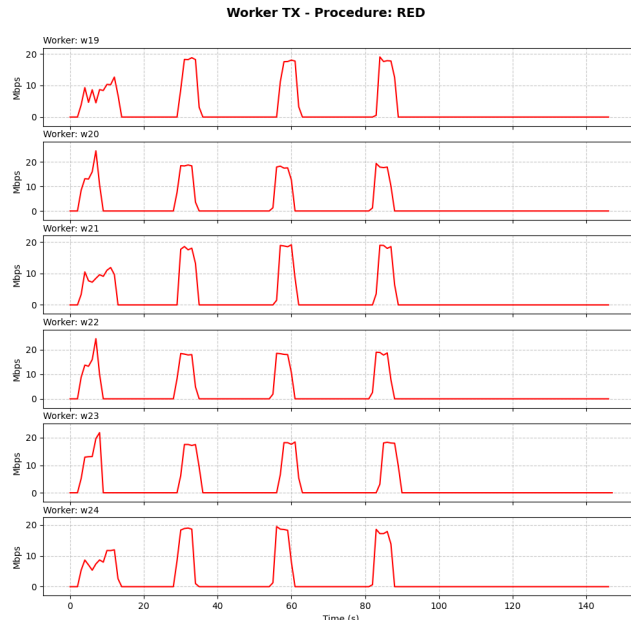


Figure 19: Default Scenario – Advanced strategy, Red result.

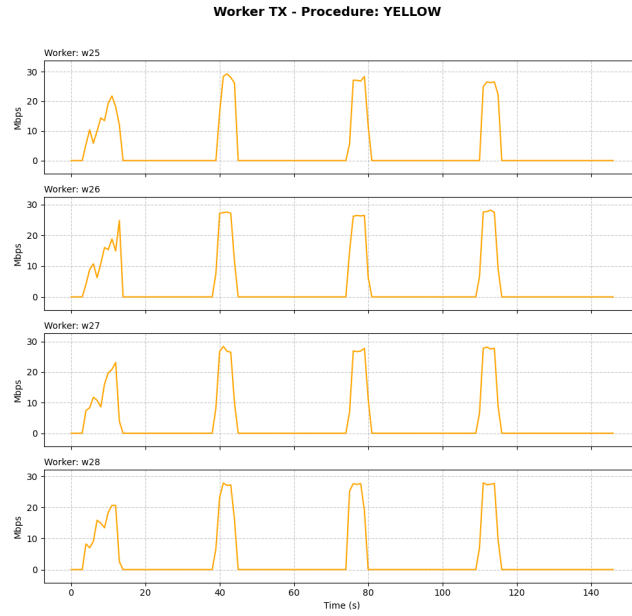


Figure 20: Default Scenario – Advanced strategy, Yellow result.

1.4 Temporal Strategy

The following plots report all available outputs for the `temporal` strategy in `default_scenario`.

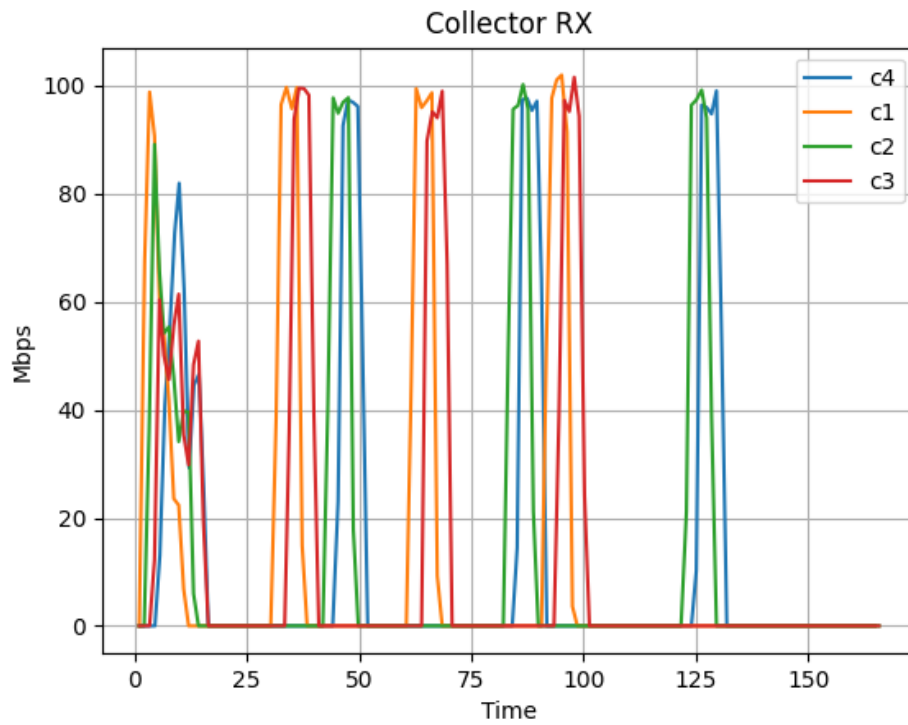


Figure 21: Default Scenario – Temporal strategy, Global result.

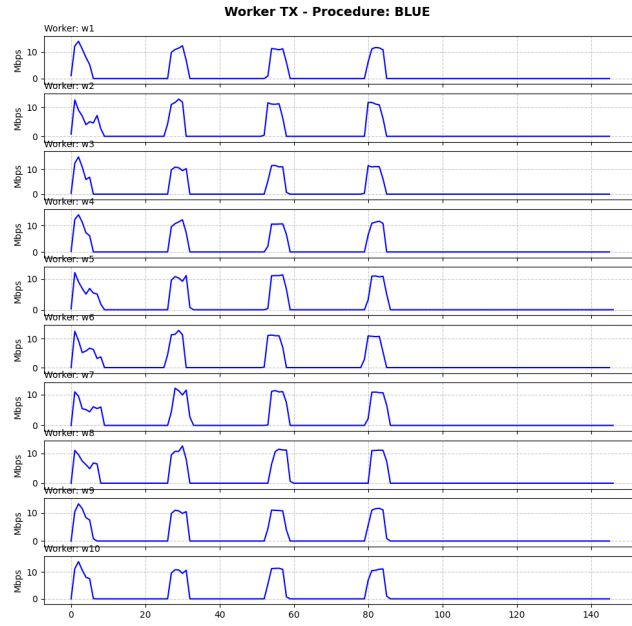


Figure 22: Default Scenario – Temporal strategy, Blue result.

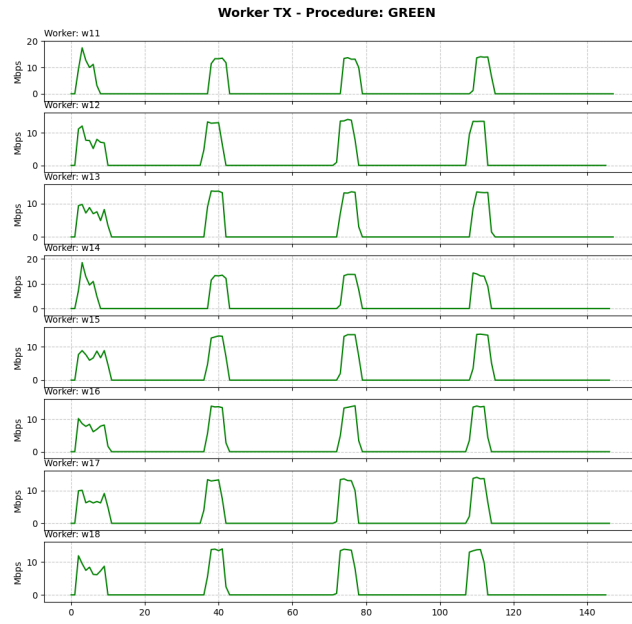


Figure 23: Default Scenario – Temporal strategy, Green result.

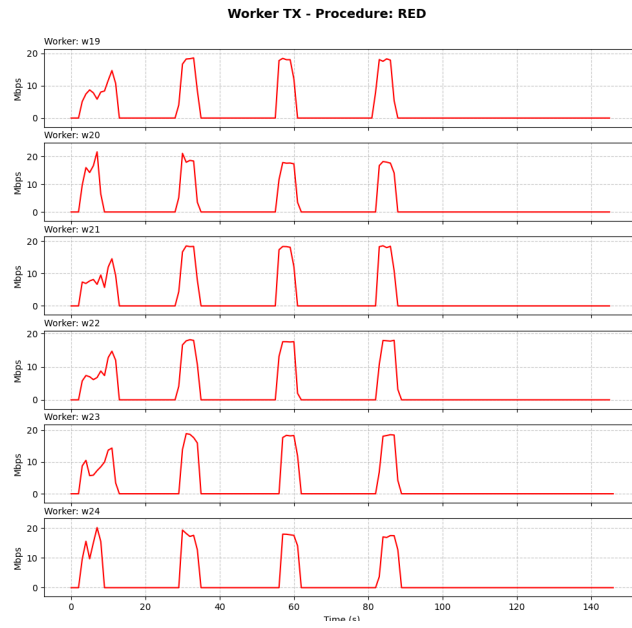


Figure 24: Default Scenario – Temporal strategy, Red result.

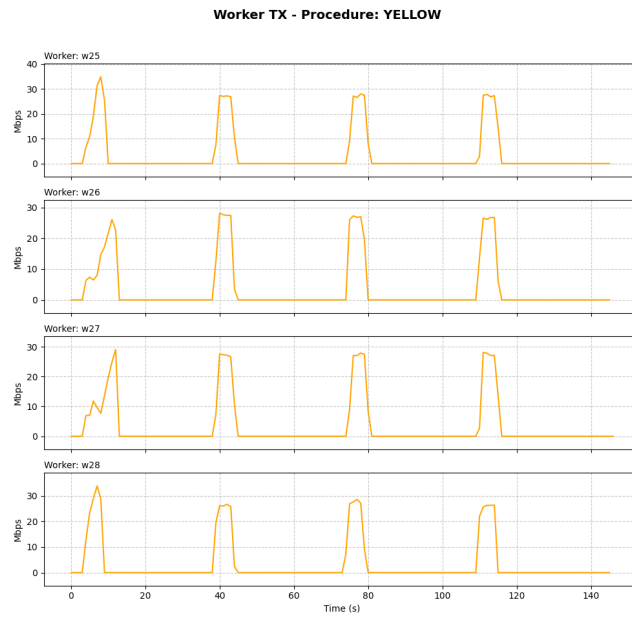


Figure 25: Default Scenario – Temporal strategy, Yellow result.

2 Scenario 1

Scenario type: `scenario_1_full_collision`. Worst-case scenario for Incast traffic. All 4 procedures start exactly at the same time ($\phi=0$) with identical cycle periods, forcing massive concurrent flows on the core fabric switches.

Training	Workers	Collector	D	T	ϕ	Cycles
Blue	7	c1 (10.0.1.1)	60.0	30	0.0	4
Green	7	c2 (10.0.1.2)	60.0	30	0.0	4
Red	7	c3 (10.0.1.3)	60.0	30	0.0	4
Yellow	7	c4 (10.0.1.4)	60.0	30	0.0	4

2.1 Simple Strategy

The following plots report all available outputs for the `simple` strategy in `scenario_1`.

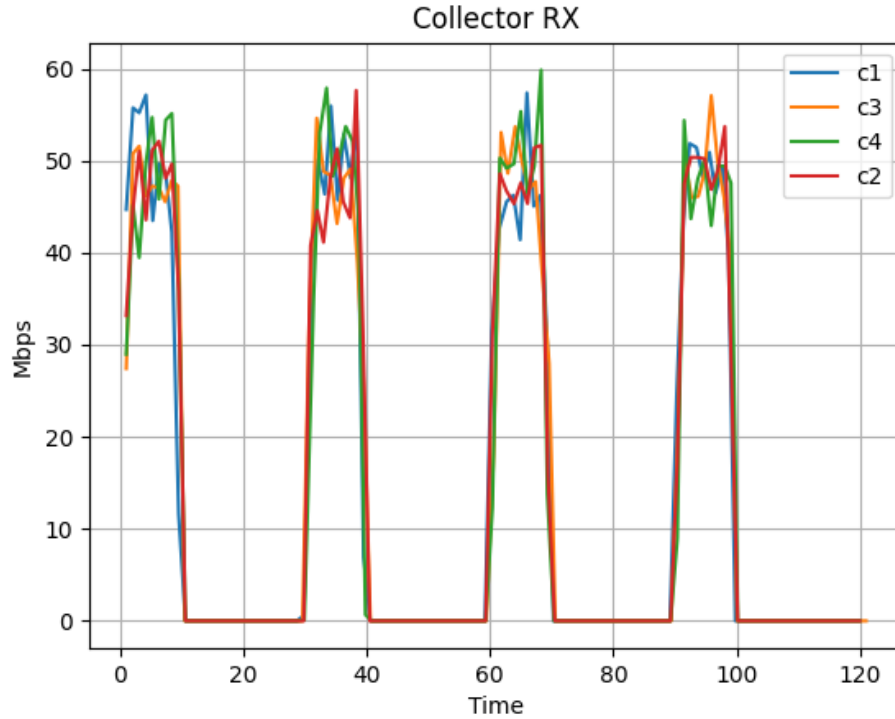


Figure 26: Scenario 1 – Simple strategy, Global result.

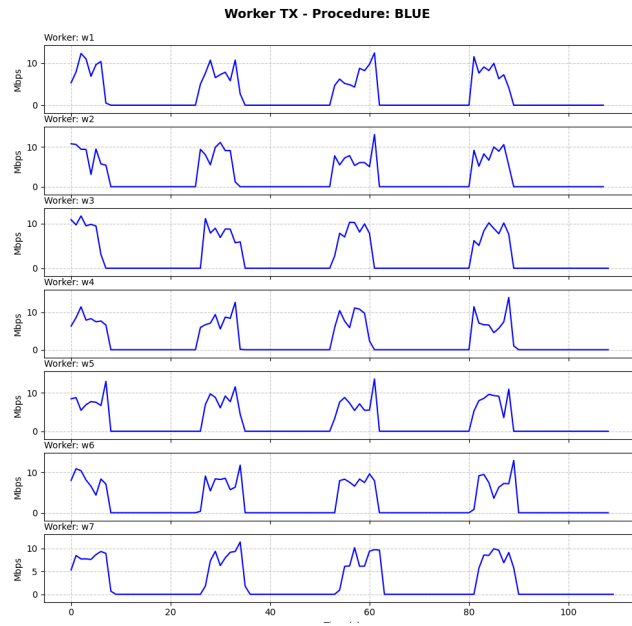


Figure 27: Scenario 1 – Simple strategy, Blue result.

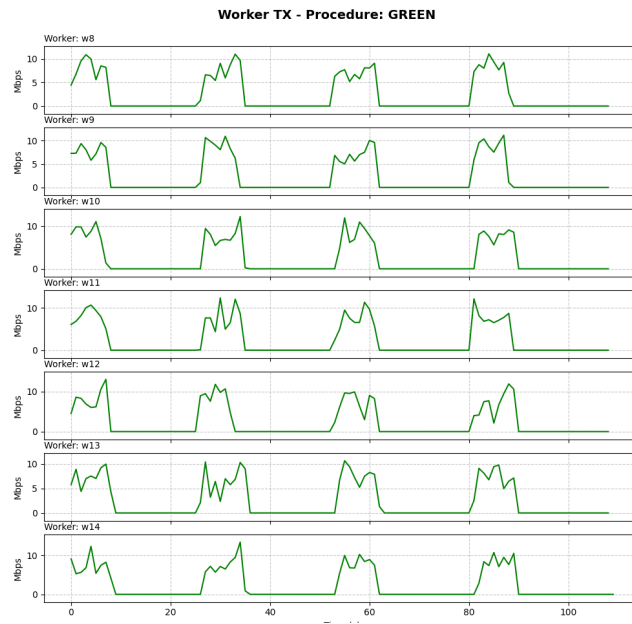


Figure 28: Scenario 1 – Simple strategy, Green result.

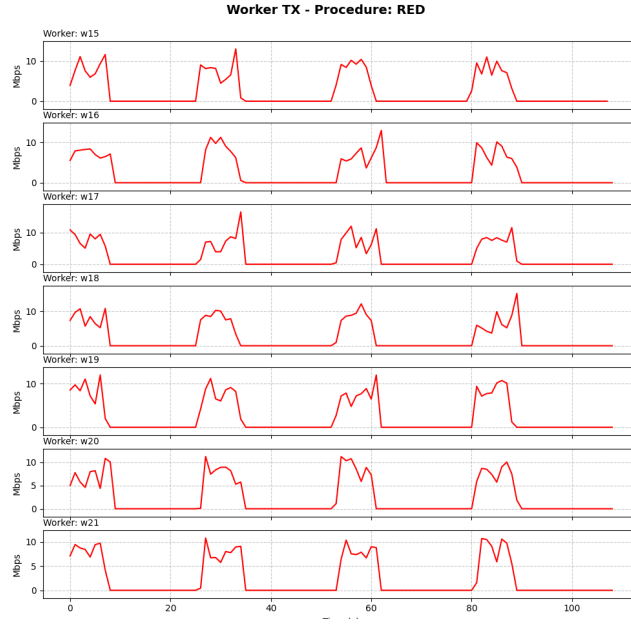


Figure 29: Scenario 1 – Simple strategy, Red result.

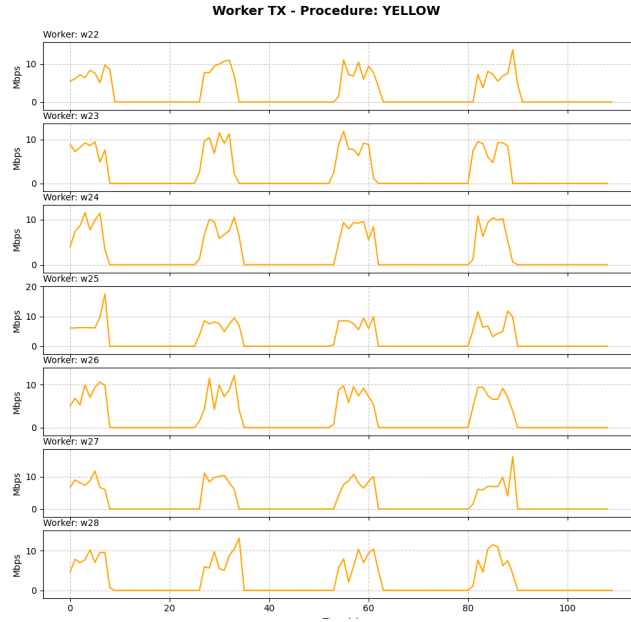


Figure 30: Scenario 1 – Simple strategy, Yellow result.

2.2 Normal Strategy

The following plots report all available outputs for the **normal** strategy in **scenario_1**.

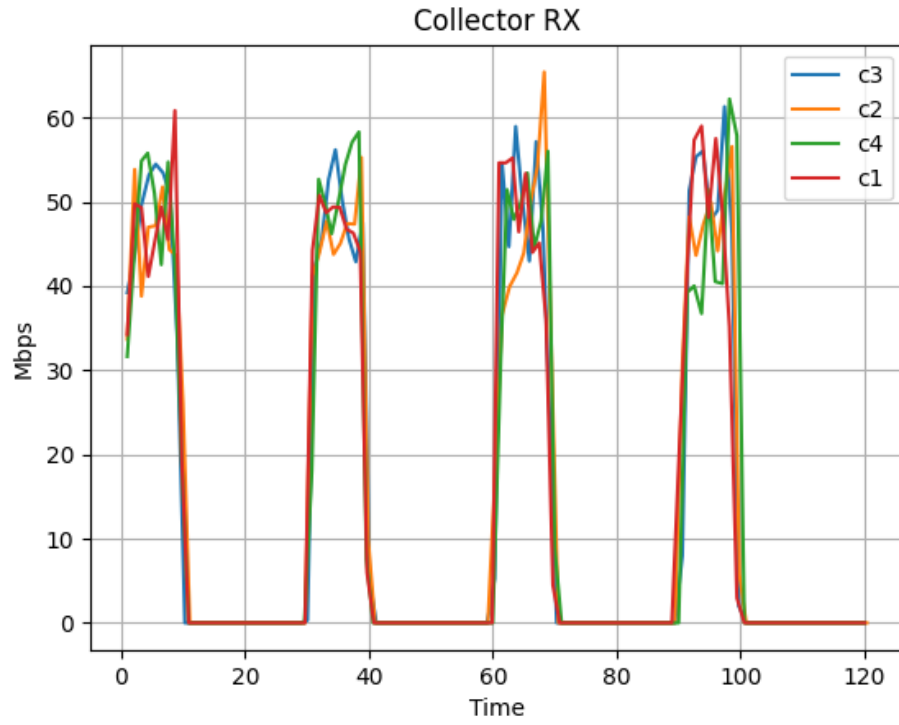


Figure 31: Scenario 1 – Normal strategy, Global result.

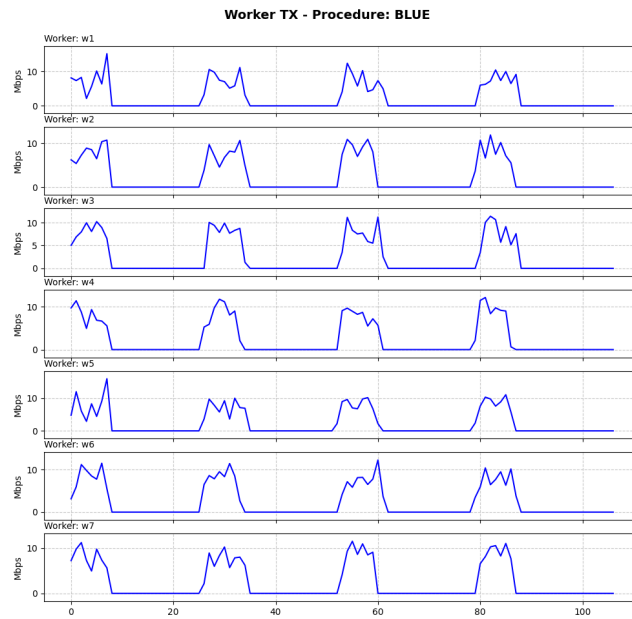


Figure 32: Scenario 1 – Normal strategy, Blue result.

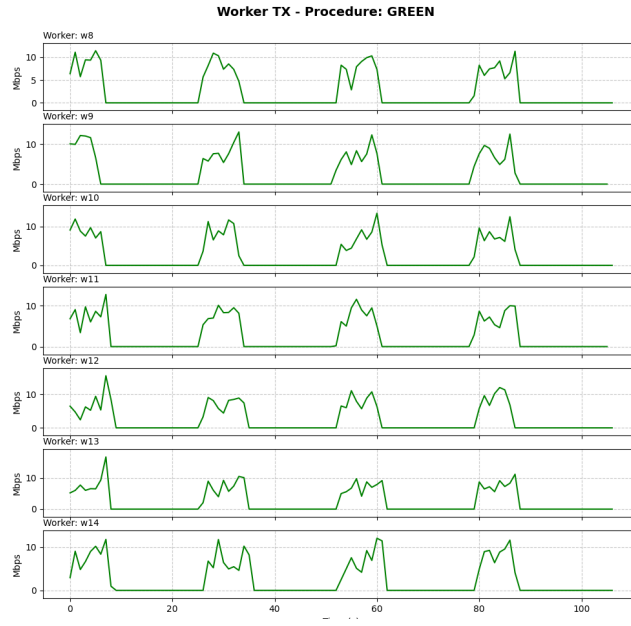


Figure 33: Scenario 1 – Normal strategy, Green result.

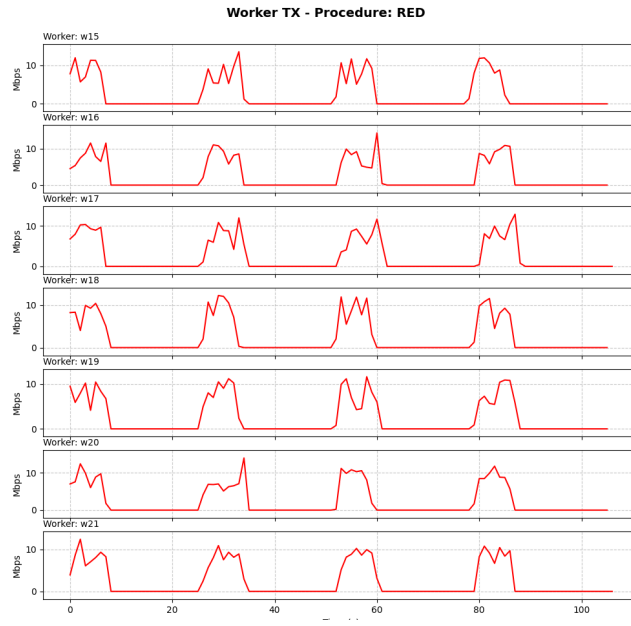


Figure 34: Scenario 1 – Normal strategy, Red result.

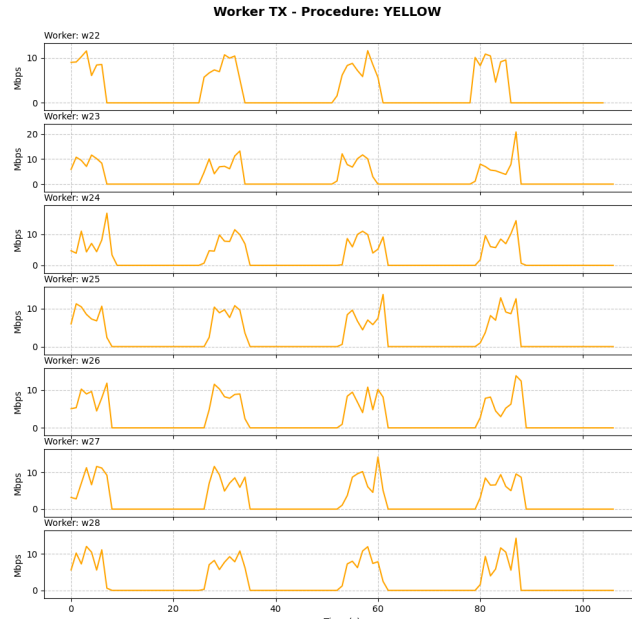


Figure 35: Scenario 1 – Normal strategy, Yellow result.

2.3 Advanced Strategy

The following plots report all available outputs for the advanced strategy in `scenario_1`.

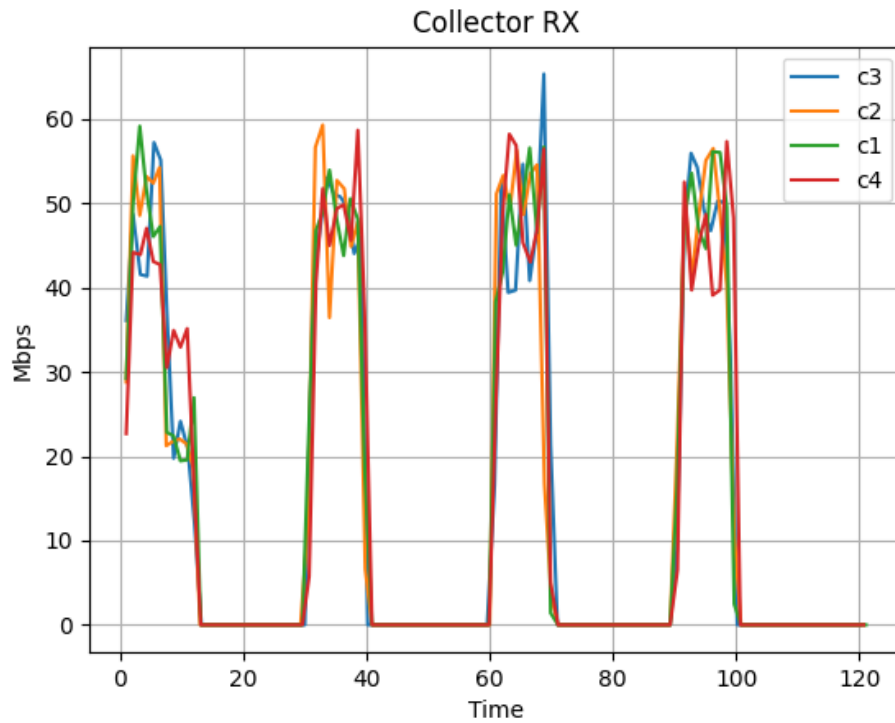


Figure 36: Scenario 1 – Advanced strategy, Global result.

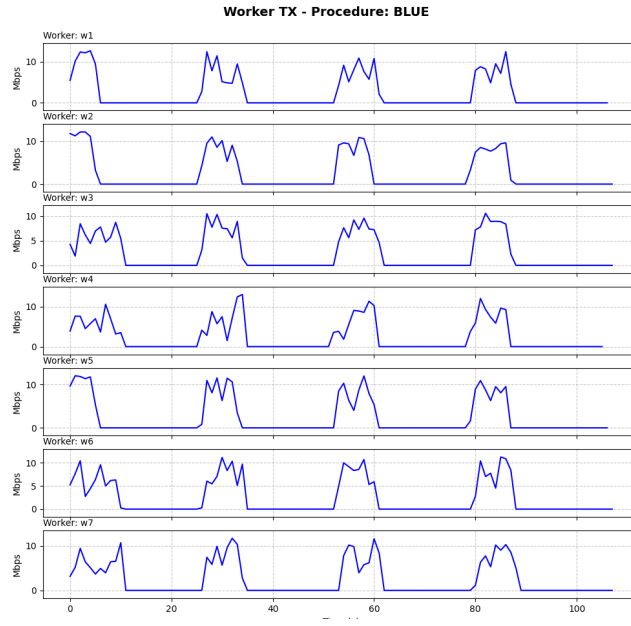


Figure 37: Scenario 1 – Advanced strategy, Blue result.

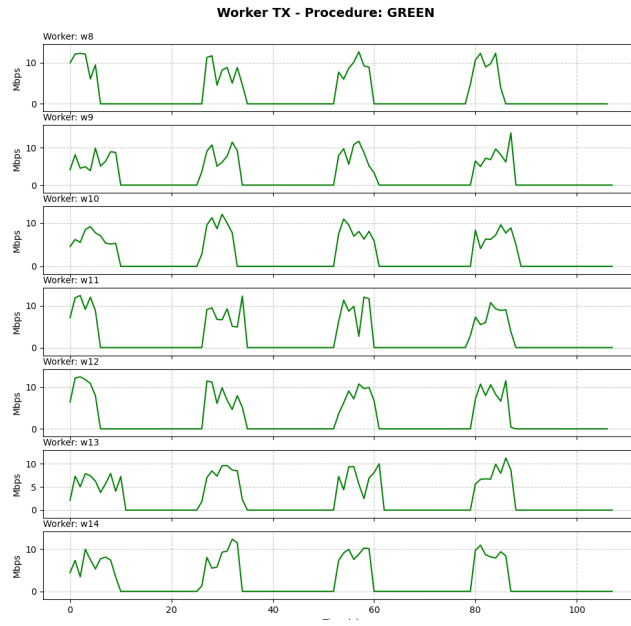


Figure 38: Scenario 1 – Advanced strategy, Green result.

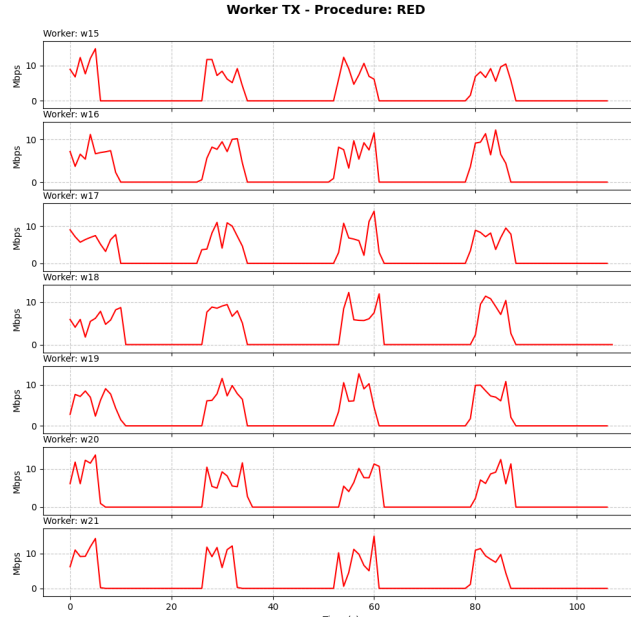


Figure 39: Scenario 1 – Advanced strategy, Red result.

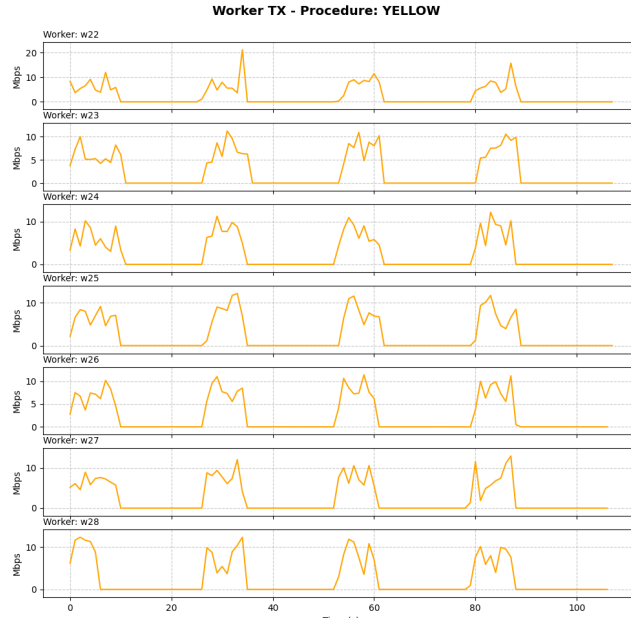


Figure 40: Scenario 1 – Advanced strategy, Yellow result.

2.4 Temporal Strategy

The following plots report all available outputs for the `temporal` strategy in `scenario_1`.

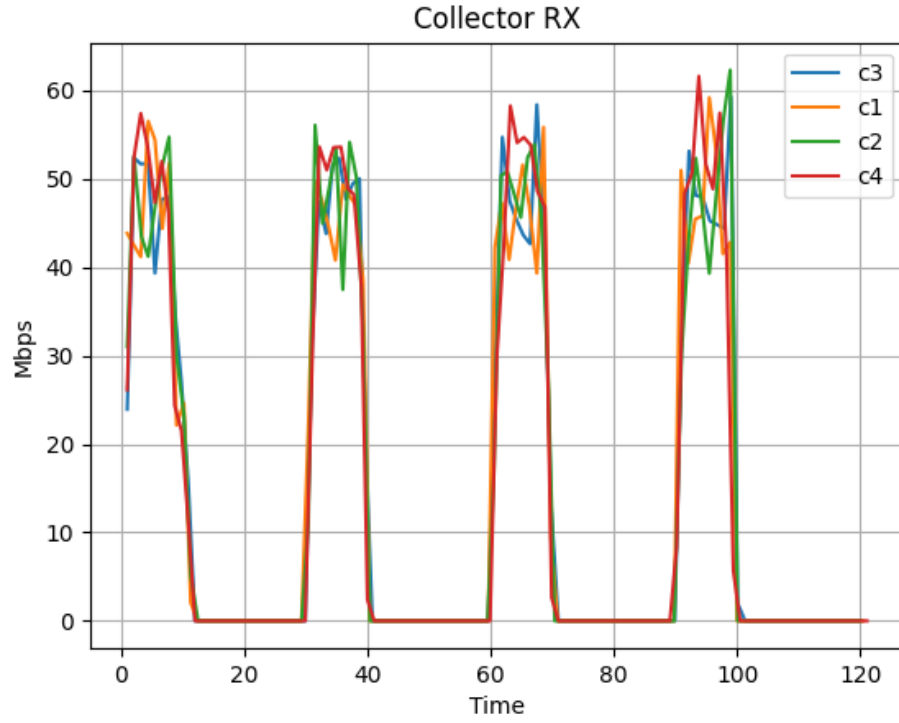


Figure 41: Scenario 1 – Temporal strategy, Global result.

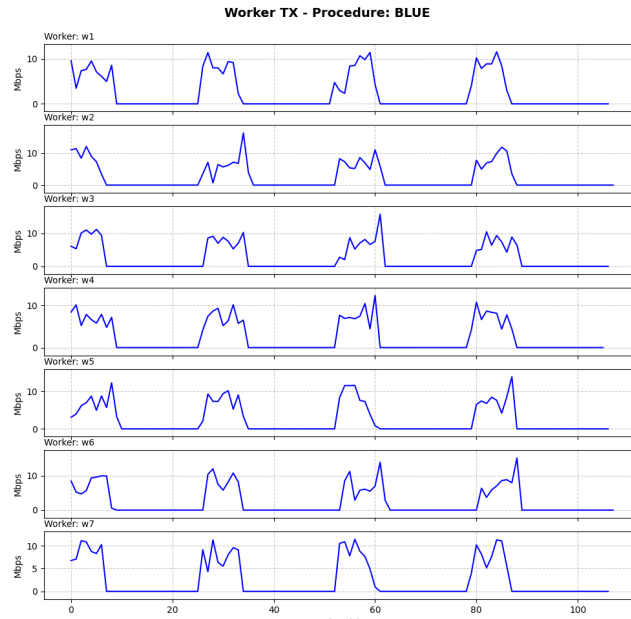


Figure 42: Scenario 1 – Temporal strategy, Blue result.

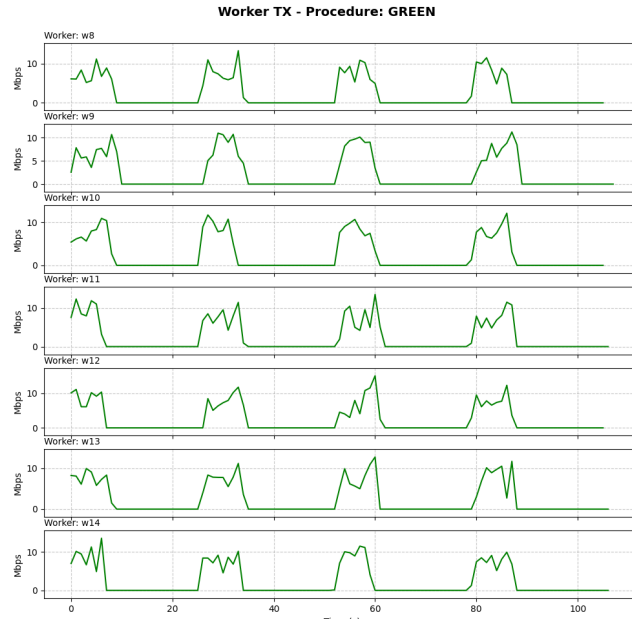


Figure 43: Scenario 1 – Temporal strategy, Green result.

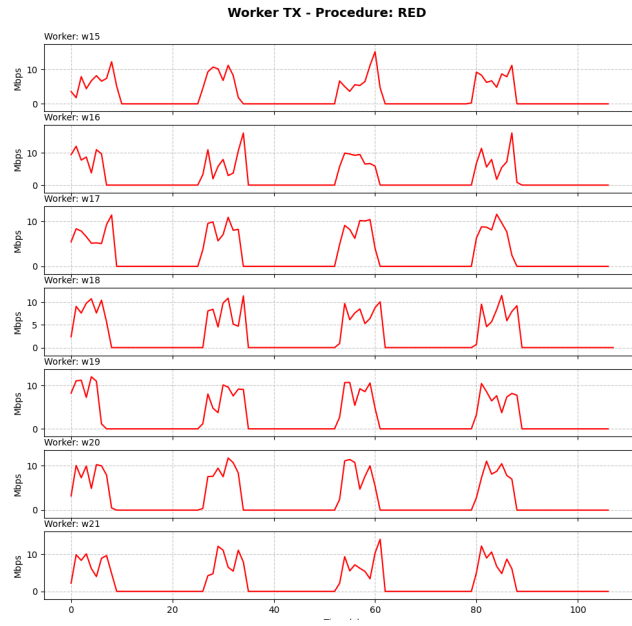


Figure 44: Scenario 1 – Temporal strategy, Red result.

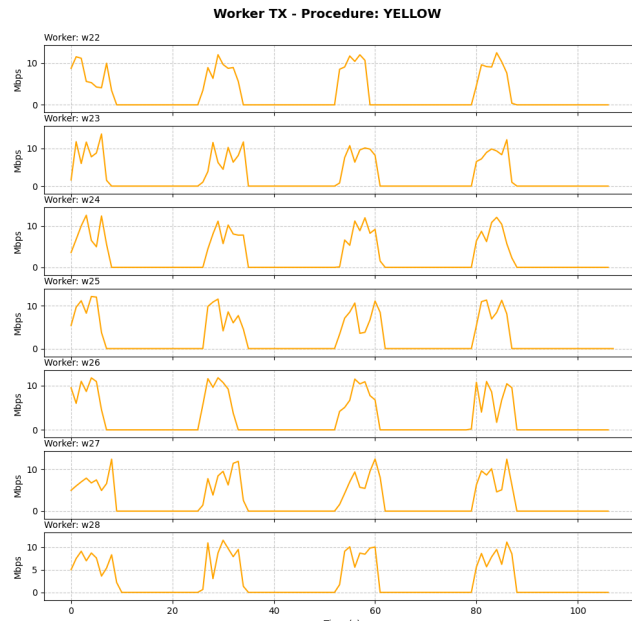


Figure 45: Scenario 1 – Temporal strategy, Yellow result.

3 Scenario 2

Scenario type: `scenario_2_asymmetric_load`. Simulates a production cluster where a massive background job (e.g., LLM training with 16 workers) runs alongside small, frequent, and short-burst jobs. Tests max-min fairness to prevent light jobs from starvation.

Training	Workers	Collector	D	T	ϕ	Cycles
Blue	16	c1 (10.0.1.1)	150.0	60	1.0	2
Green	4	c2 (10.0.1.2)	30.0	20	5.0	5
Red	4	c3 (10.0.1.3)	30.0	20	10.0	5
Yellow	4	c4 (10.0.1.4)	30.0	20	15.0	5

3.1 Simple Strategy

The following plots report all available outputs for the `simple` strategy in `scenario_2`.

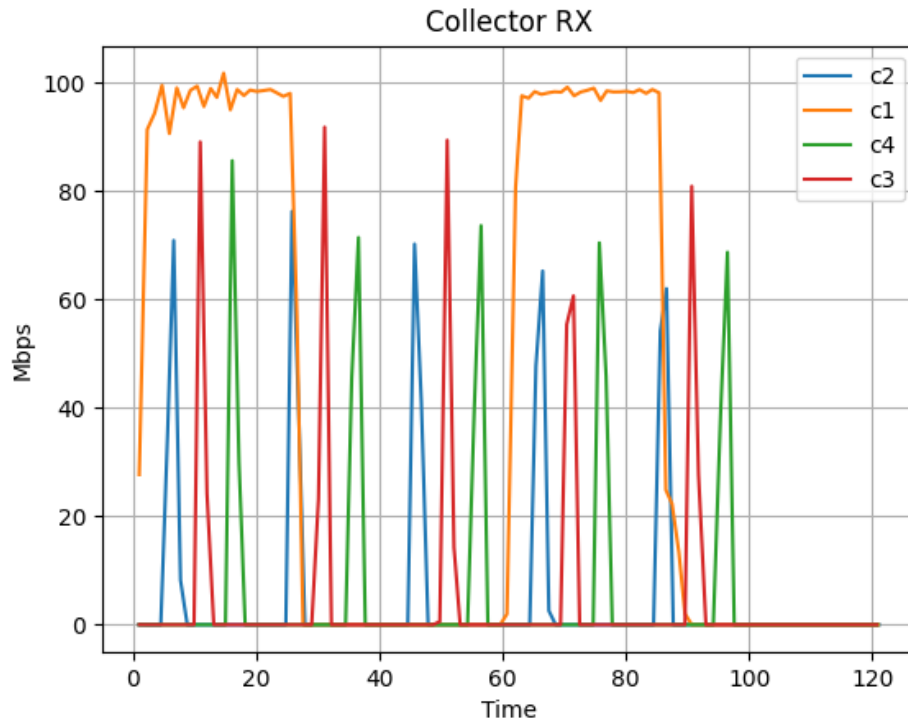


Figure 46: Scenario 2 – Simple strategy, Global result.

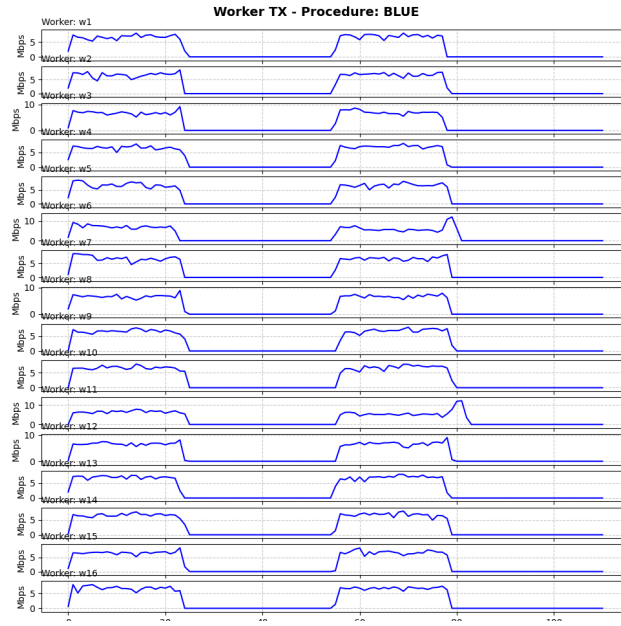


Figure 47: Scenario 2 – Simple strategy, Blue result.

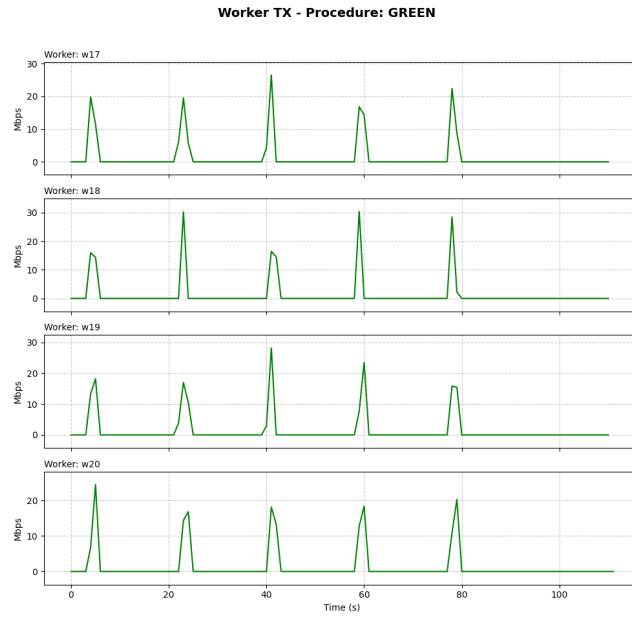


Figure 48: Scenario 2 – Simple strategy, Green result.

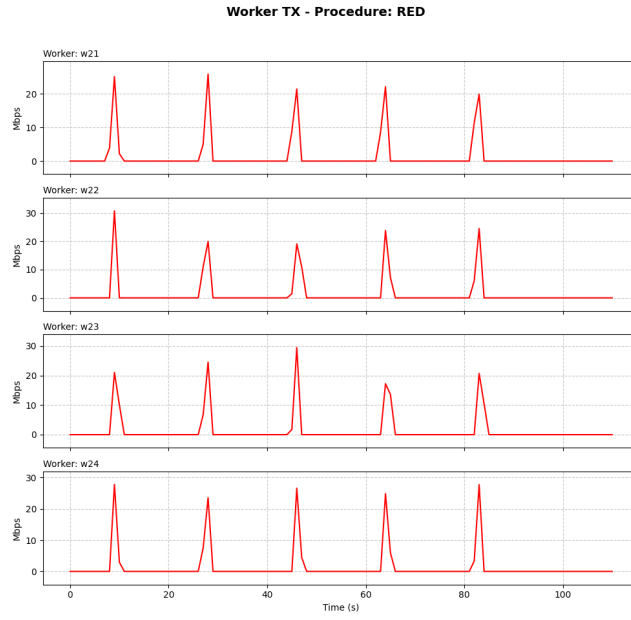


Figure 49: Scenario 2 – Simple strategy, Red result.

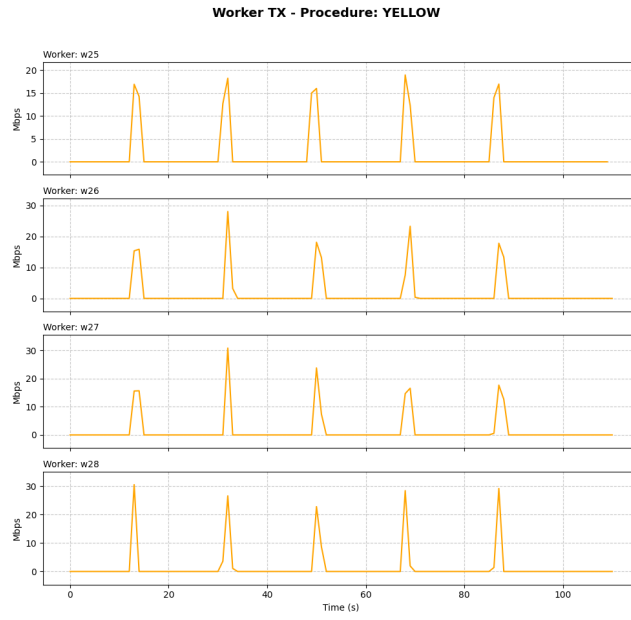


Figure 50: Scenario 2 – Simple strategy, Yellow result.

3.2 Normal Strategy

The following plots report all available outputs for the `normal` strategy in `scenario_2`.

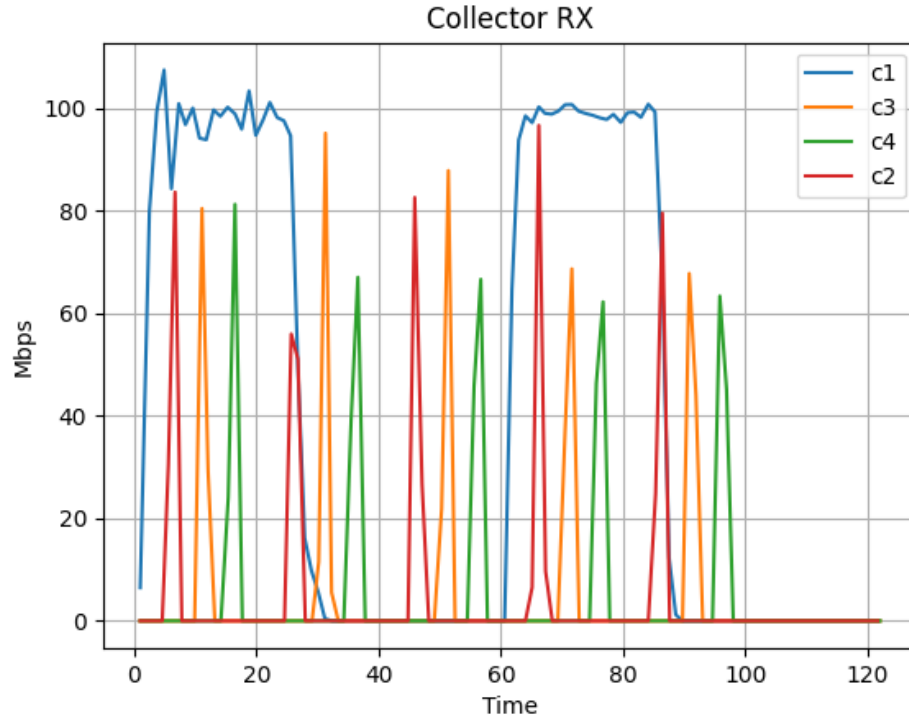


Figure 51: Scenario 2 – Normal strategy, Global result.

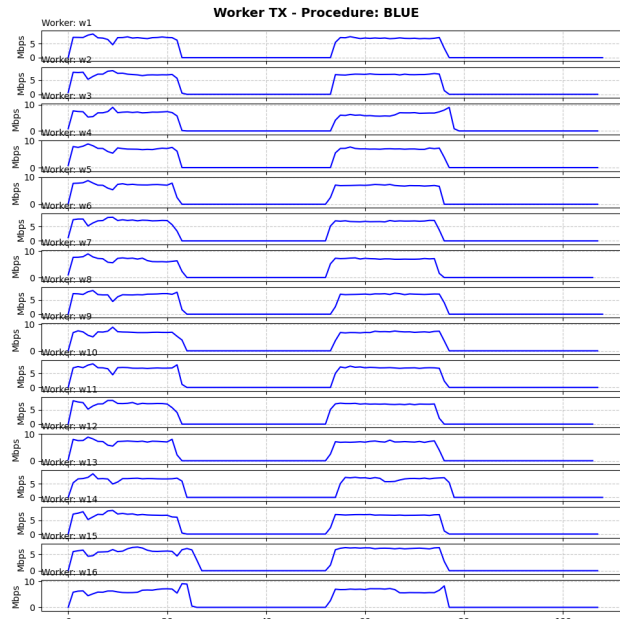


Figure 52: Scenario 2 – Normal strategy, Blue result.

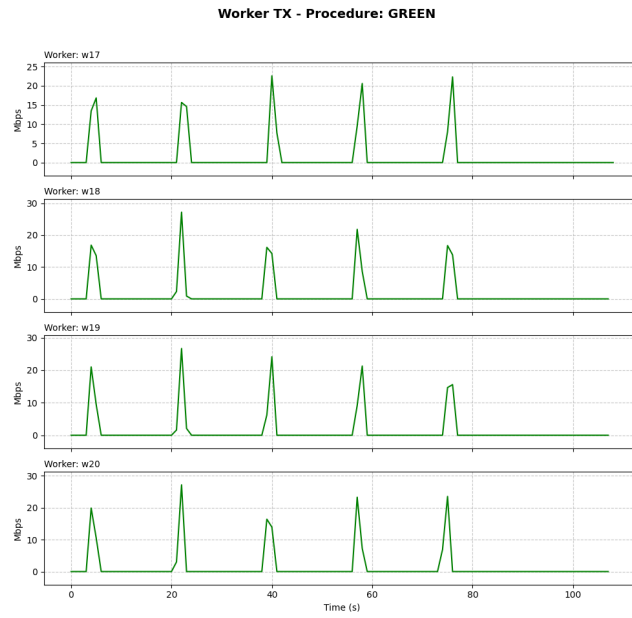


Figure 53: Scenario 2 – Normal strategy, Green result.

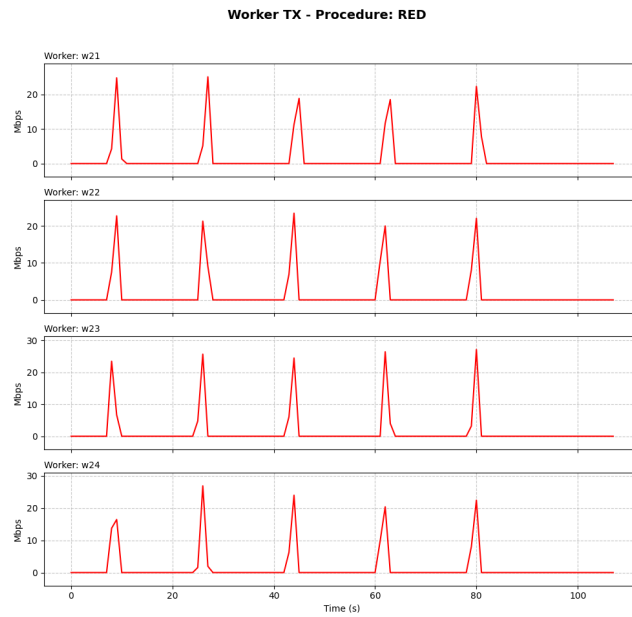


Figure 54: Scenario 2 – Normal strategy, Red result.

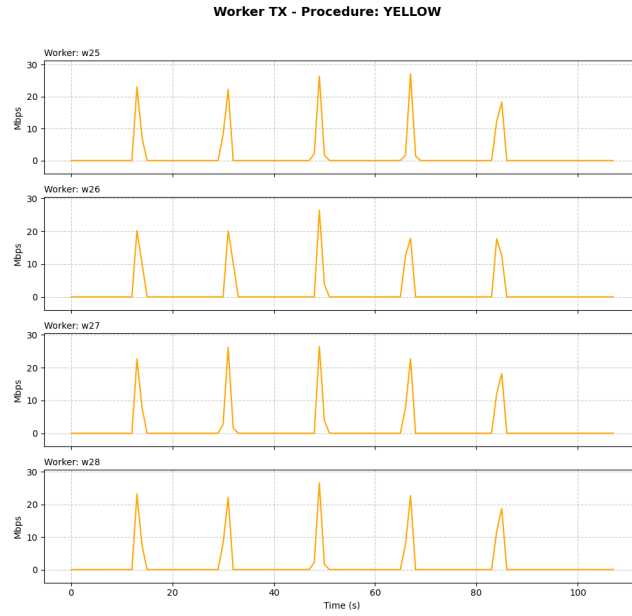


Figure 55: Scenario 2 – Normal strategy, Yellow result.

3.3 Advanced Strategy

The following plots report all available outputs for the advanced strategy in `scenario_2`.

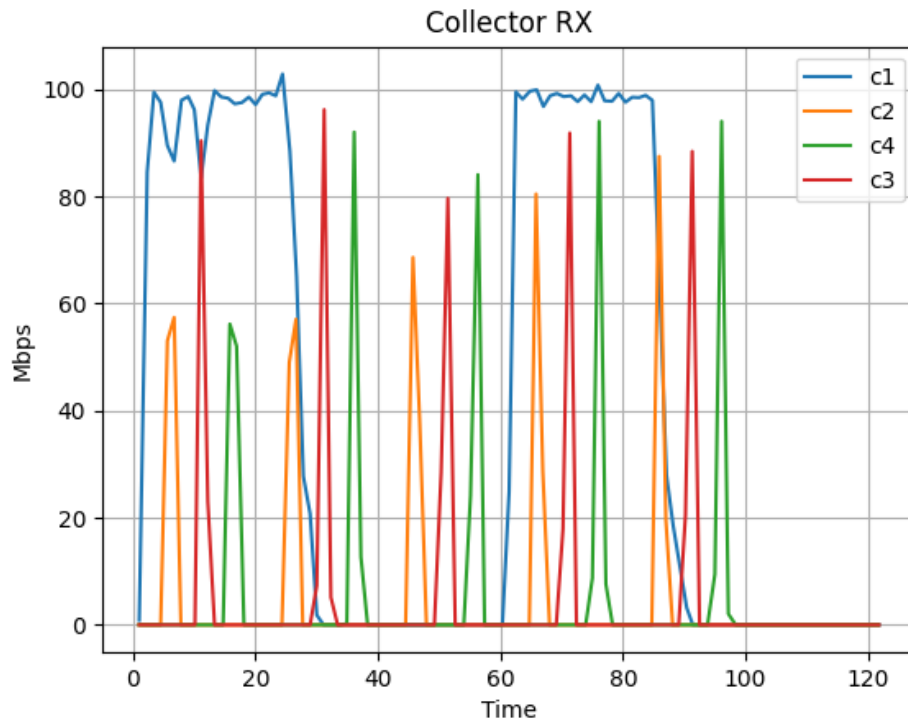


Figure 56: Scenario 2 – Advanced strategy, Global result.

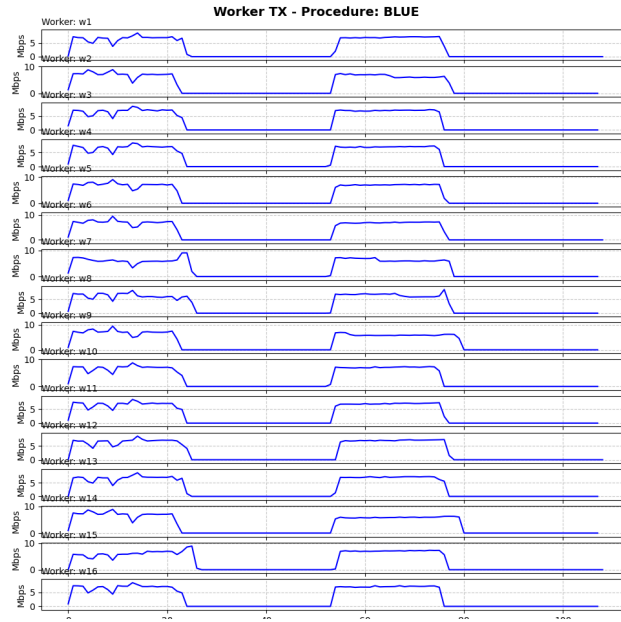


Figure 57: Scenario 2 – Advanced strategy, Blue result.

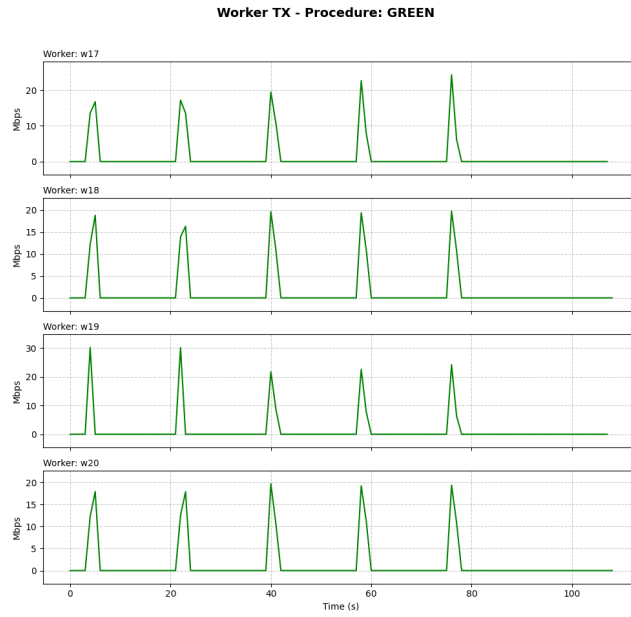


Figure 58: Scenario 2 – Advanced strategy, Green result.

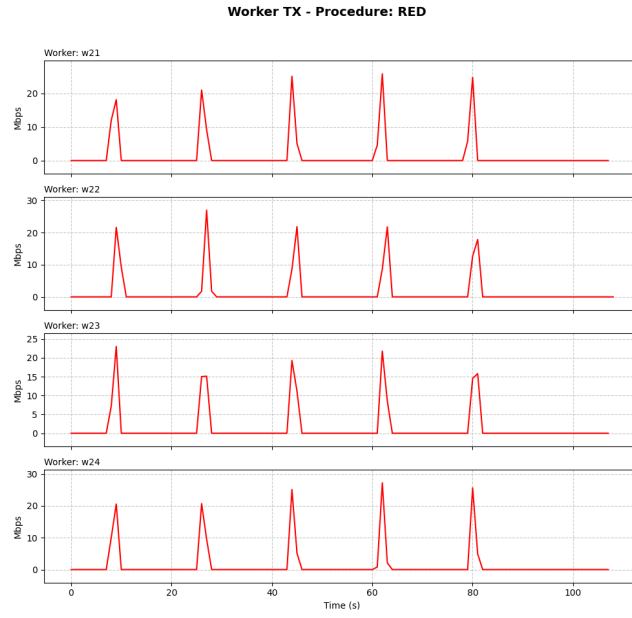


Figure 59: Scenario 2 – Advanced strategy, Red result.

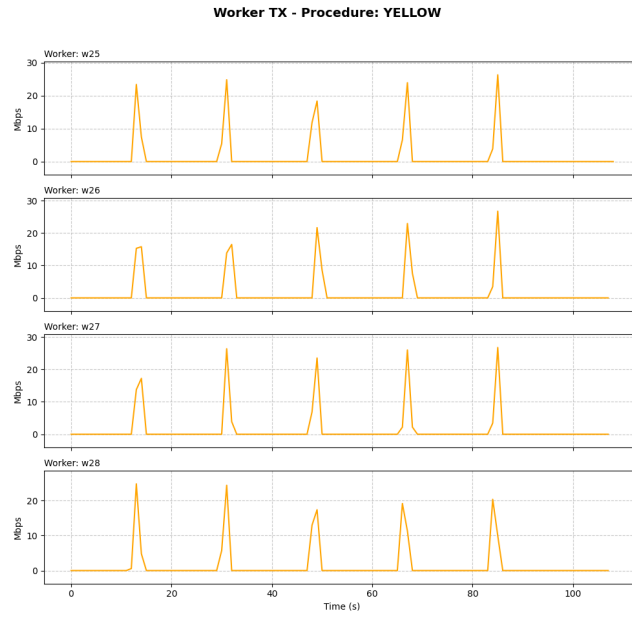


Figure 60: Scenario 2 – Advanced strategy, Yellow result.

3.4 Temporal Strategy

The following plots report all available outputs for the `temporal` strategy in `scenario_2`.

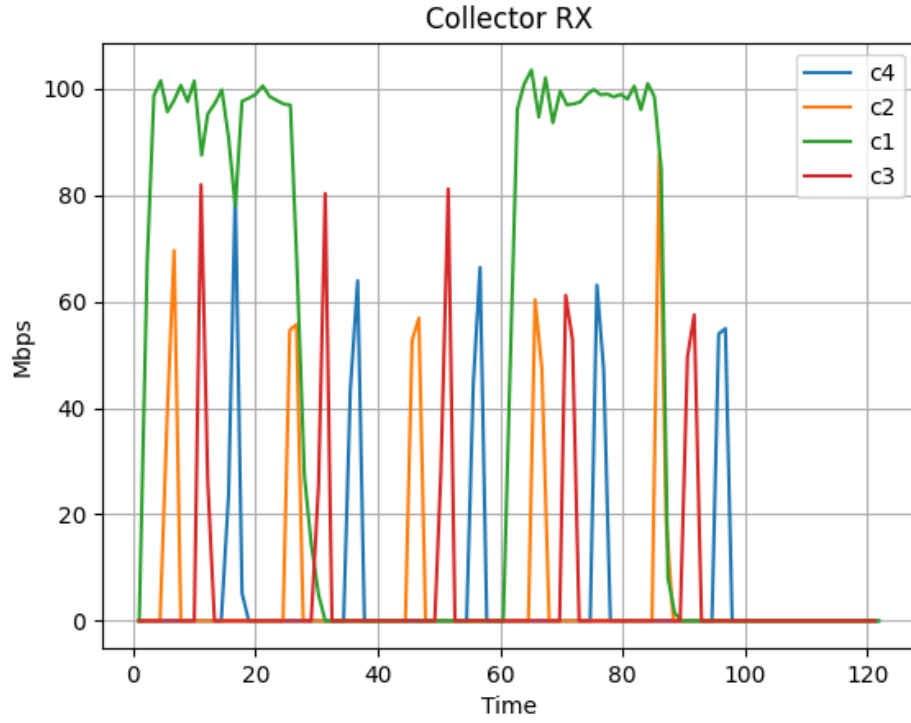


Figure 61: Scenario 2 – Temporal strategy, Global result.

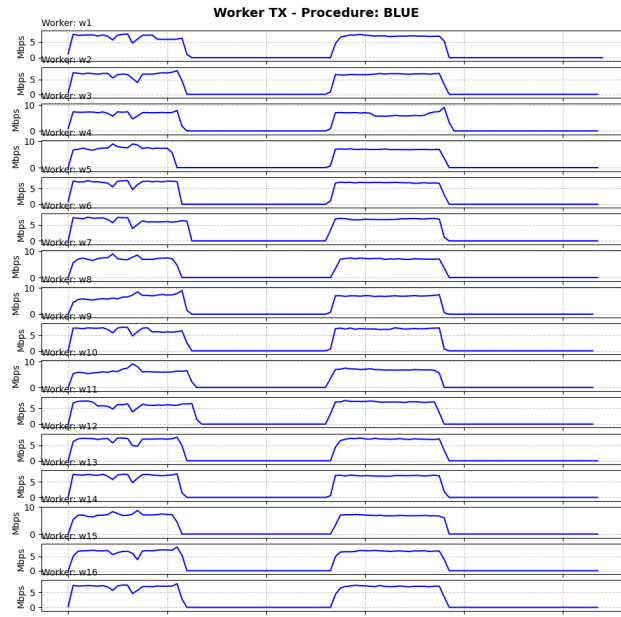


Figure 62: Scenario 2 – Temporal strategy, Blue result.

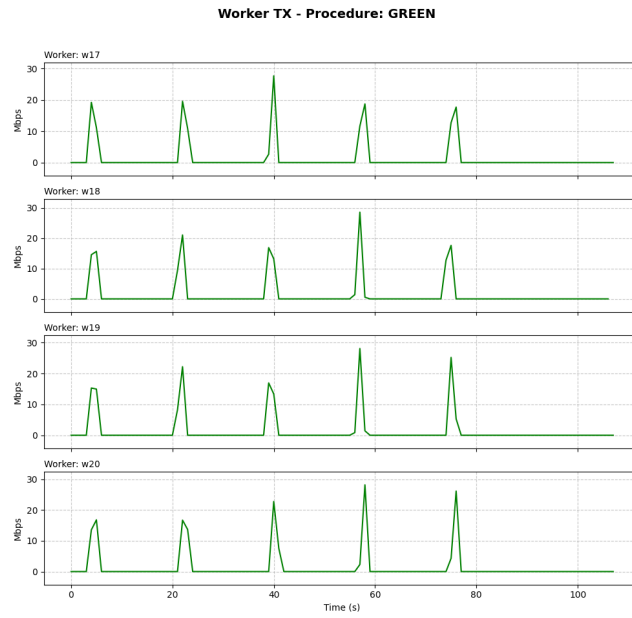


Figure 63: Scenario 2 – Temporal strategy, Green result.

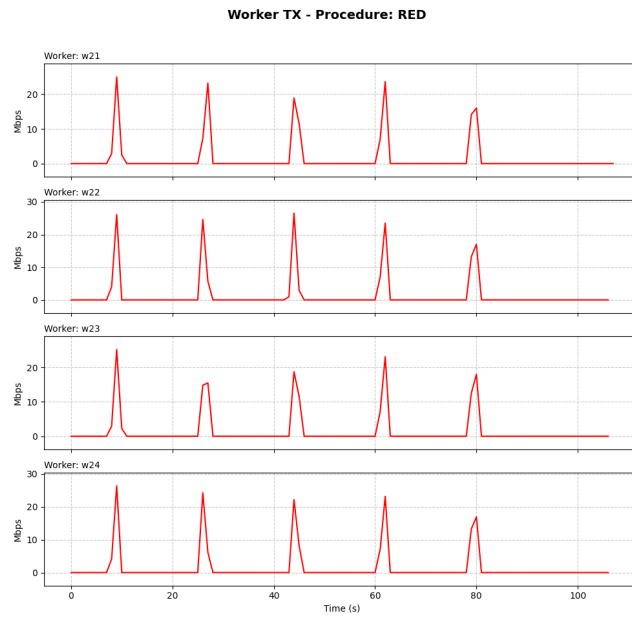


Figure 64: Scenario 2 – Temporal strategy, Red result.

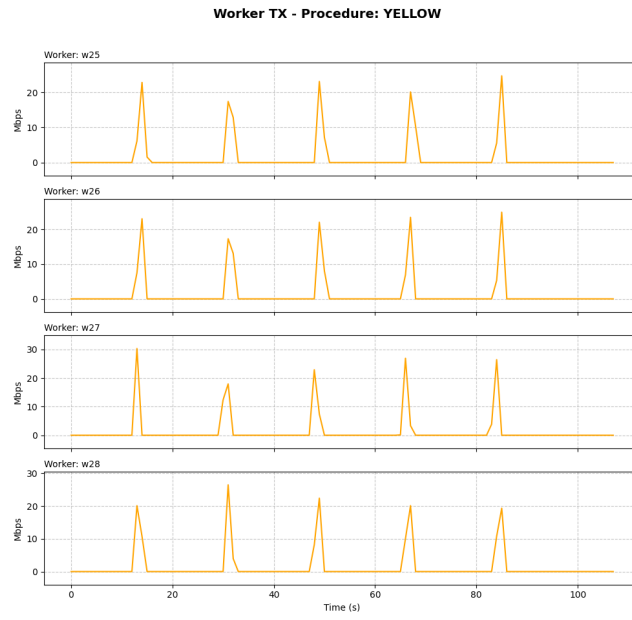


Figure 65: Scenario 2 – Temporal strategy, Yellow result.

4 Scenario 3

Scenario type: `scenario_3_stochastic_cascade`. Creates a continuous, staggered flow of distributed training cycles. Phase offsets (ϕ) are carefully shifted so that network buffers are constantly utilized, evaluating the controller’s dynamic path re-allocation stability over time.

Training	Workers	Collector	D	T	ϕ	Cycles
Blue	7	c1 (10.0.1.1)	70.0	25	0.0	5
Green	7	c2 (10.0.1.2)	70.0	25	6.0	5
Red	7	c3 (10.0.1.3)	70.0	25	12.0	5
Yellow	7	c4 (10.0.1.4)	70.0	25	18.0	5

4.1 Simple Strategy

The following plots report all available outputs for the `simple` strategy in `scenario_3`.

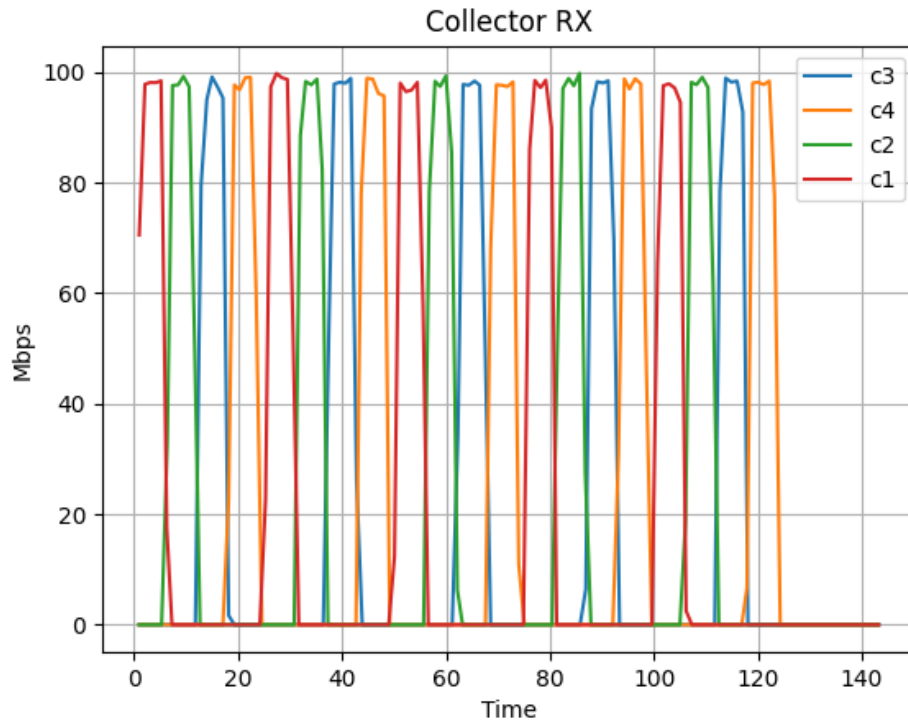


Figure 66: Scenario 3 – Simple strategy, Global result.

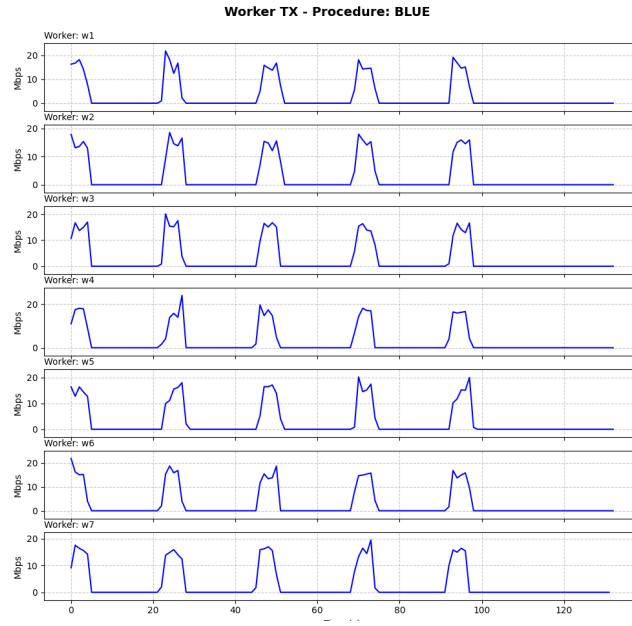


Figure 67: Scenario 3 – Simple strategy, Blue result.

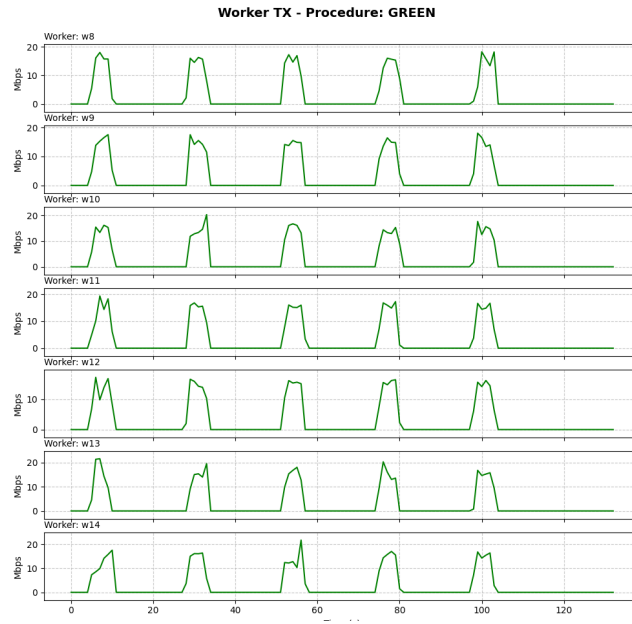


Figure 68: Scenario 3 – Simple strategy, Green result.

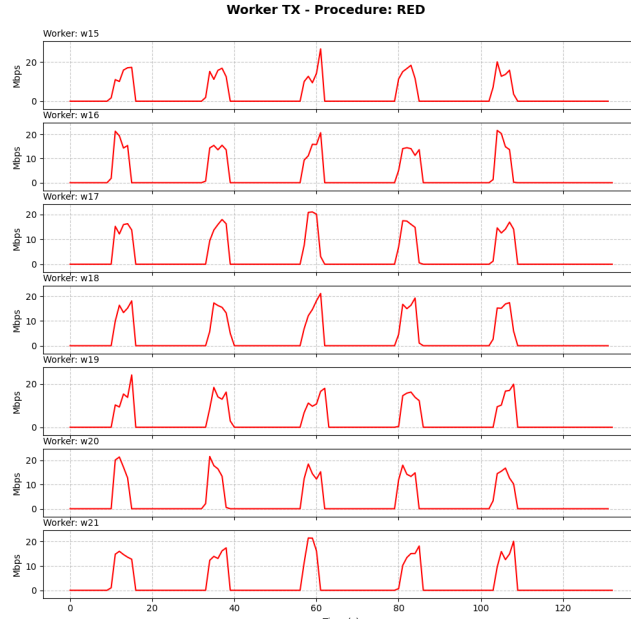


Figure 69: Scenario 3 – Simple strategy, Red result.

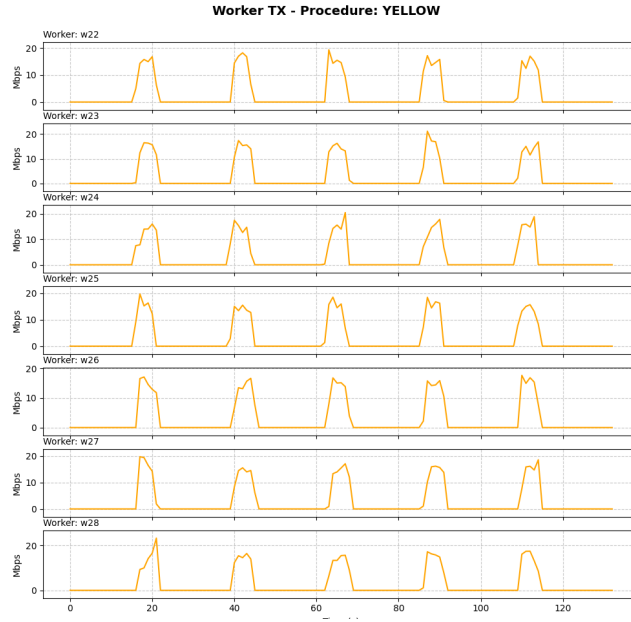


Figure 70: Scenario 3 – Simple strategy, Yellow result.

4.2 Normal Strategy

The following plots report all available outputs for the `normal` strategy in `scenario_3`.

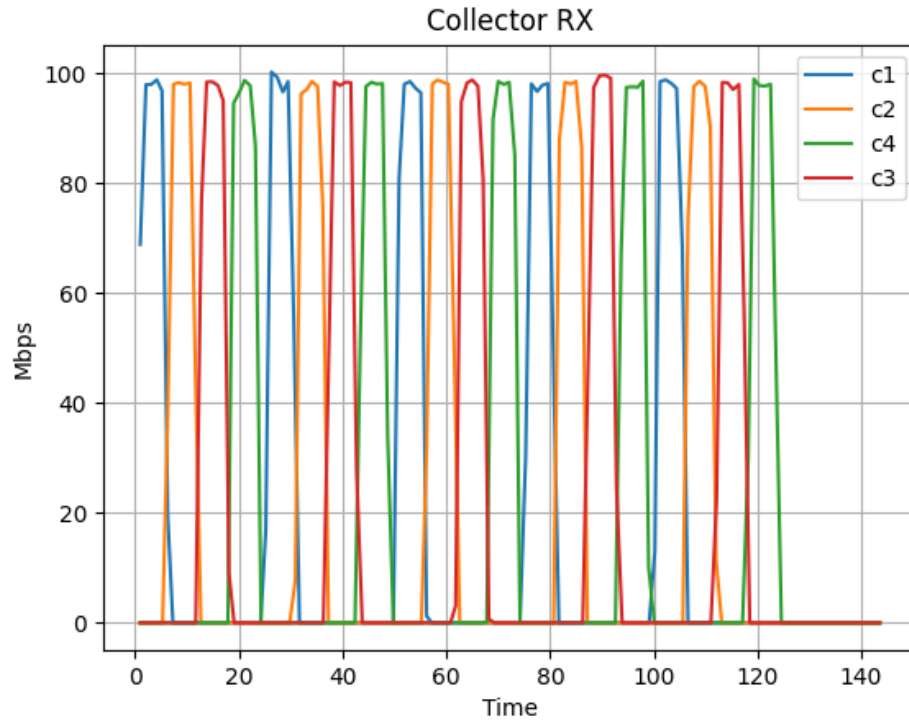


Figure 71: Scenario 3 – Normal strategy, Global result.

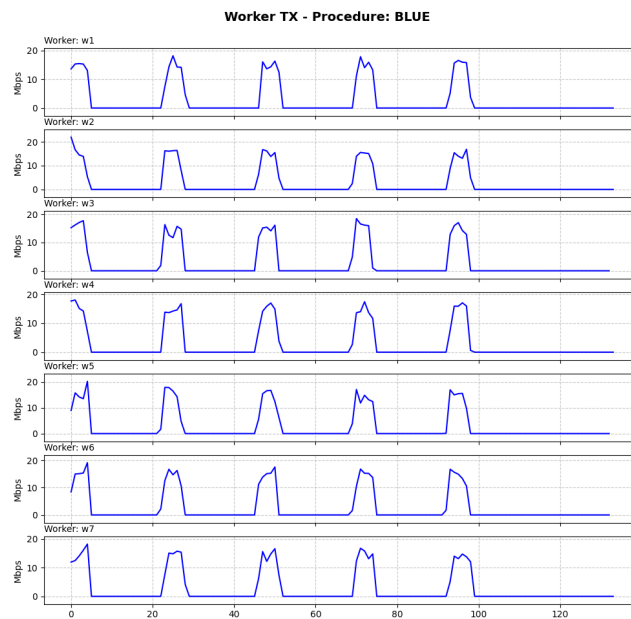


Figure 72: Scenario 3 – Normal strategy, Blue result.

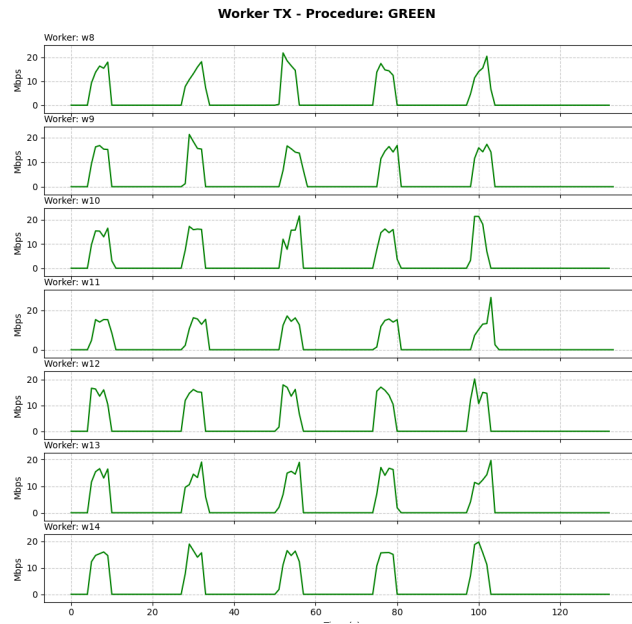


Figure 73: Scenario 3 – Normal strategy, Green result.

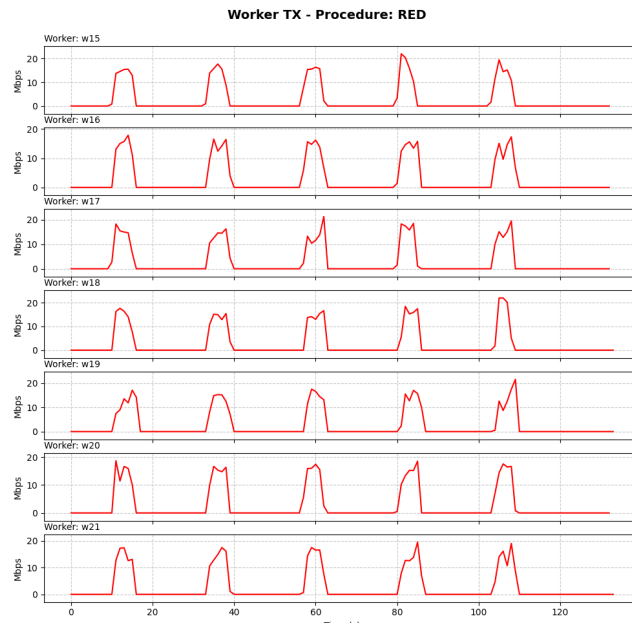


Figure 74: Scenario 3 – Normal strategy, Red result.

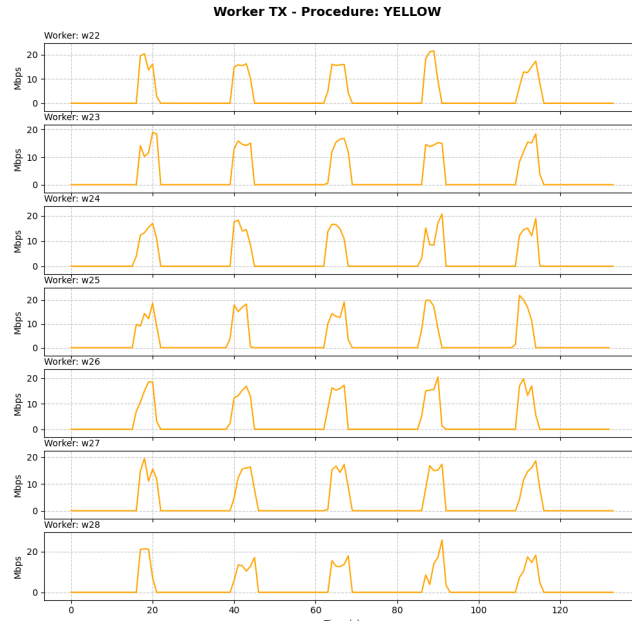


Figure 75: Scenario 3 – Normal strategy, Yellow result.

4.3 Advanced Strategy

The following plots report all available outputs for the **advanced** strategy in `scenario_3`.

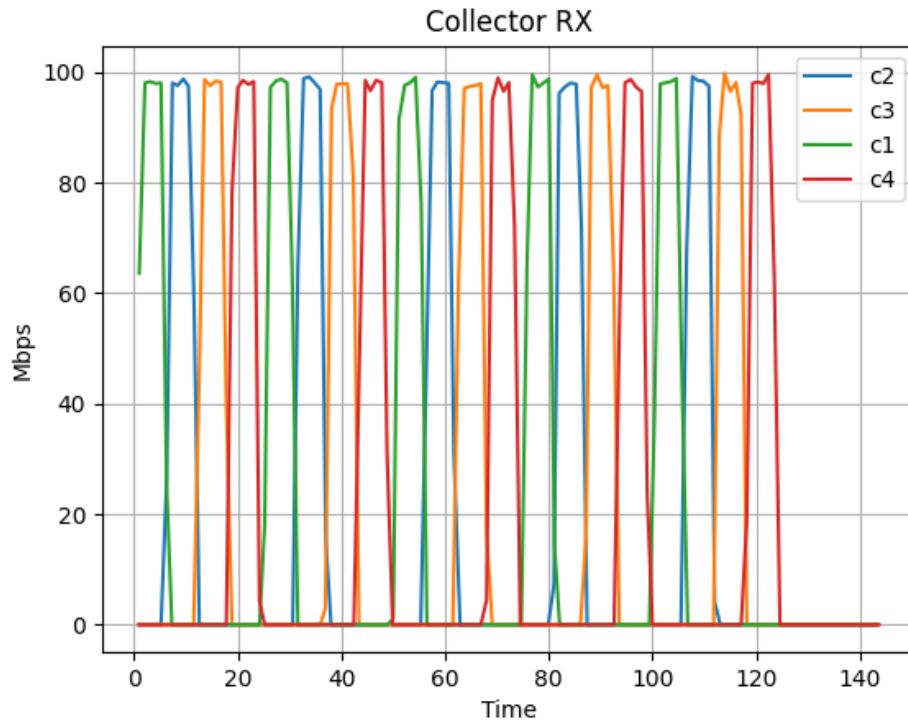


Figure 76: Scenario 3 – Advanced strategy, Global result.

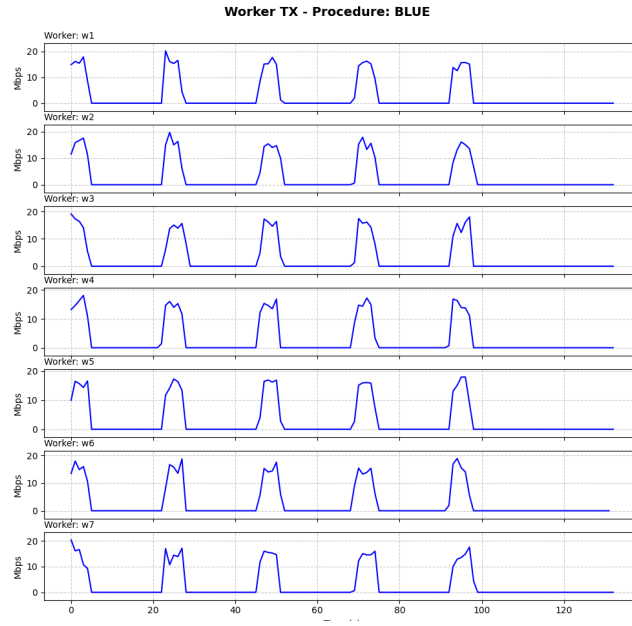


Figure 77: Scenario 3 – Advanced strategy, Blue result.

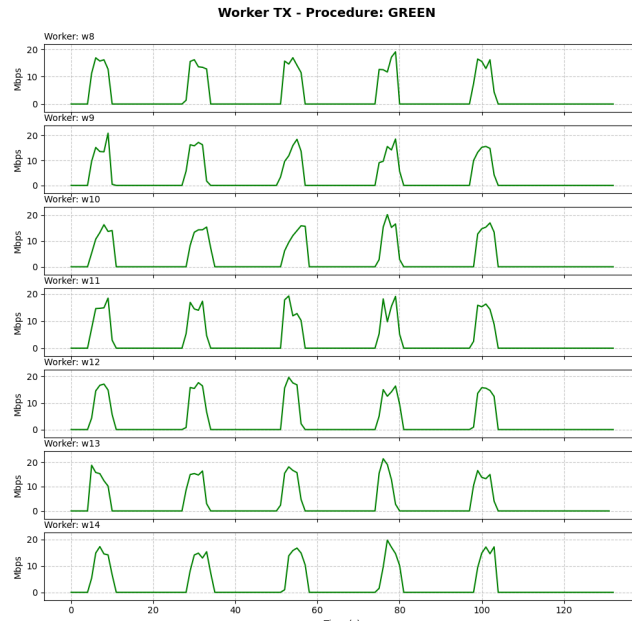


Figure 78: Scenario 3 – Advanced strategy, Green result.

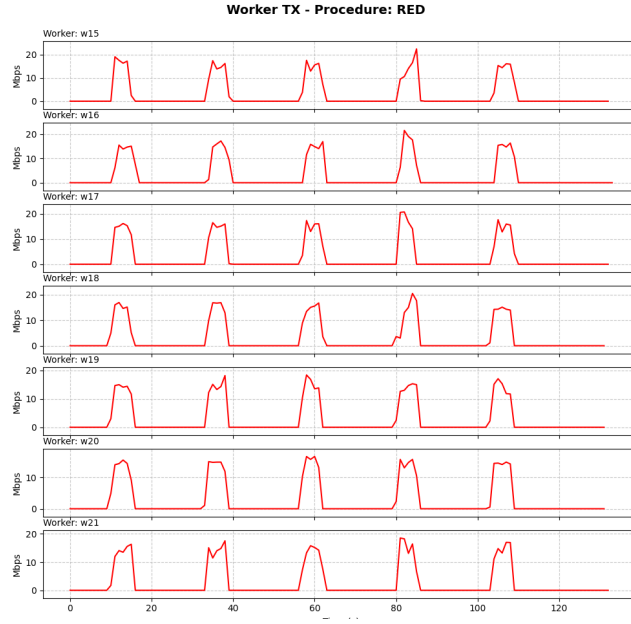


Figure 79: Scenario 3 – Advanced strategy, Red result.

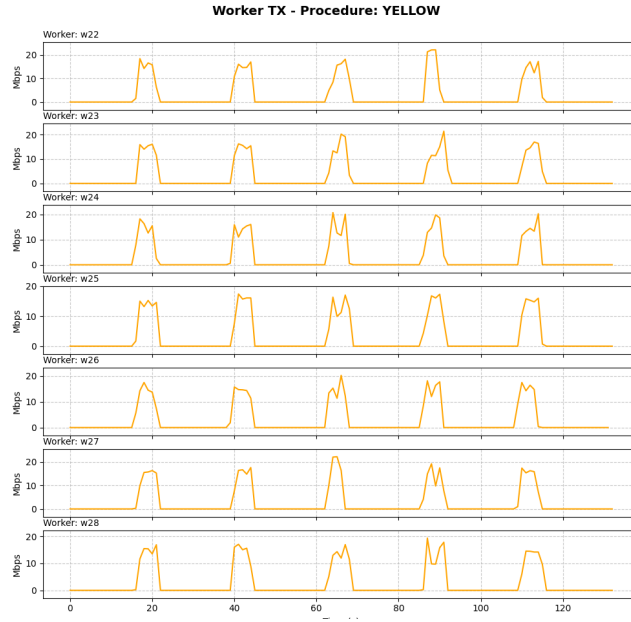


Figure 80: Scenario 3 – Advanced strategy, Yellow result.

4.4 Temporal Strategy

The following plots report all available outputs for the `temporal` strategy in `scenario_3`.

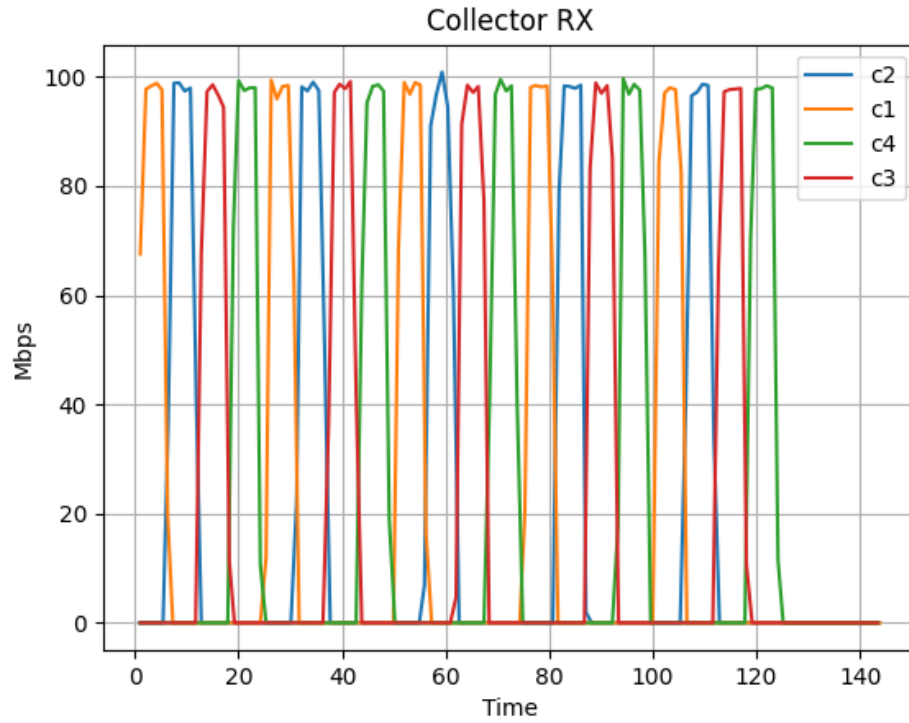


Figure 81: Scenario 3 – Temporal strategy, Global result.

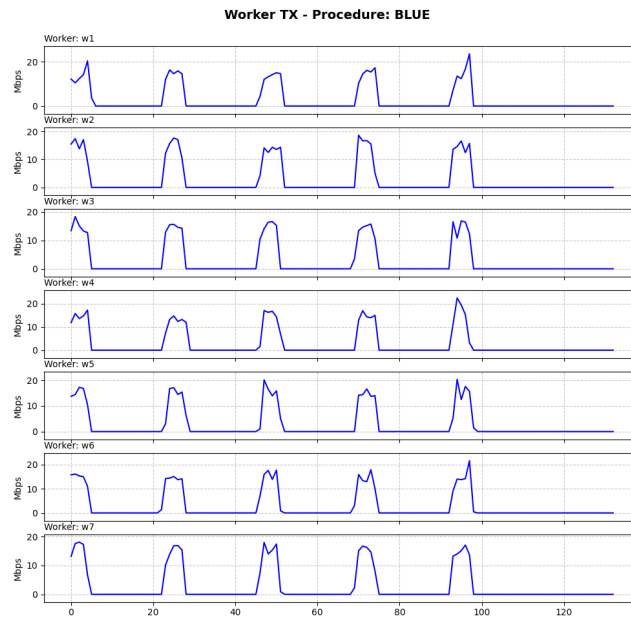


Figure 82: Scenario 3 – Temporal strategy, Blue result.

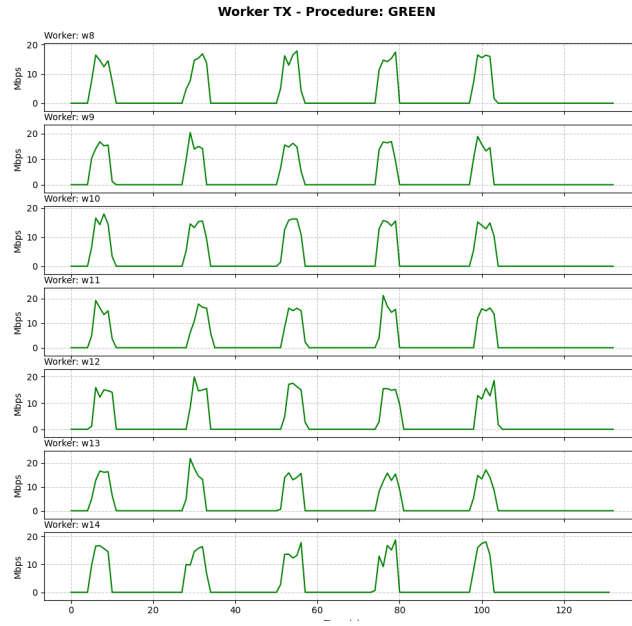


Figure 83: Scenario 3 – Temporal strategy, Green result.

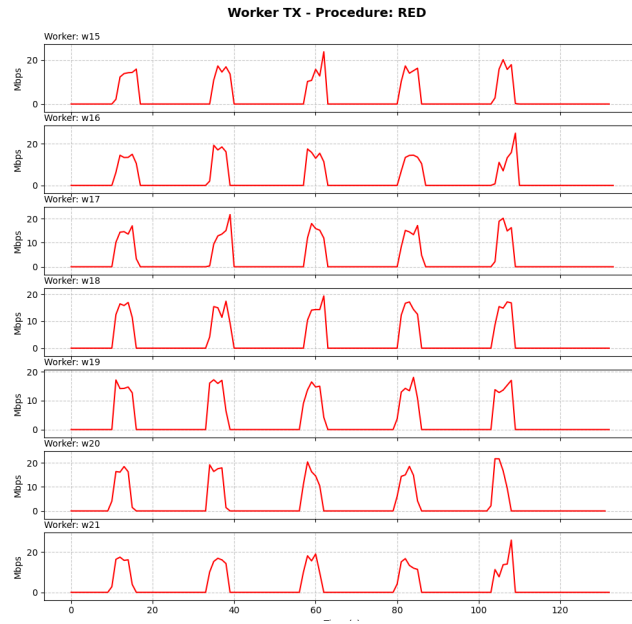


Figure 84: Scenario 3 – Temporal strategy, Red result.

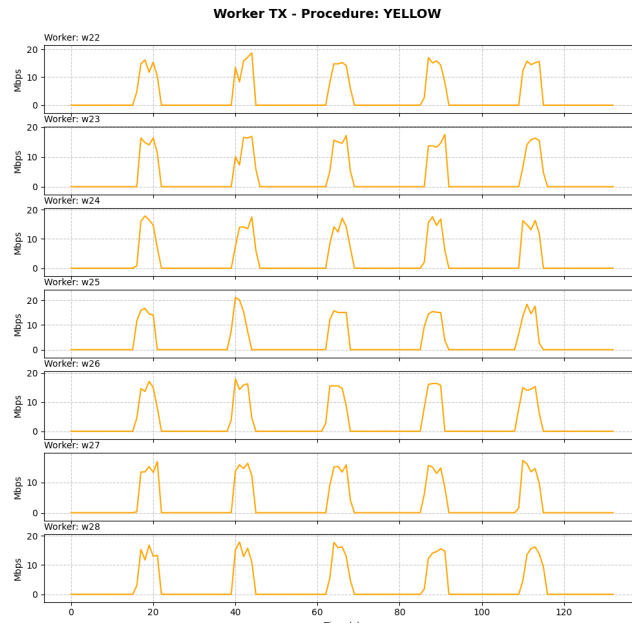


Figure 85: Scenario 3 – Temporal strategy, Yellow result.

5 Scenario 4

Scenario type: `scenario_4_zero_overlap_baseline`. A baseline control scenario where bursts are strictly isolated. Each training round begins only when the previous one has completely finished, meaning the core network never experiences cross-traffic congestion.

Training	Workers	Collector	D	T	ϕ	Cycles
Blue	7	c1 (10.0.1.1)	80.0	40	0.0	2
Green	7	c2 (10.0.1.2)	80.0	40	10.0	2
Red	7	c3 (10.0.1.3)	80.0	40	20.0	2
Yellow	7	c4 (10.0.1.4)	80.0	40	30.0	2

5.1 Simple Strategy

The following plots report all available outputs for the `simple` strategy in `scenario_4`.

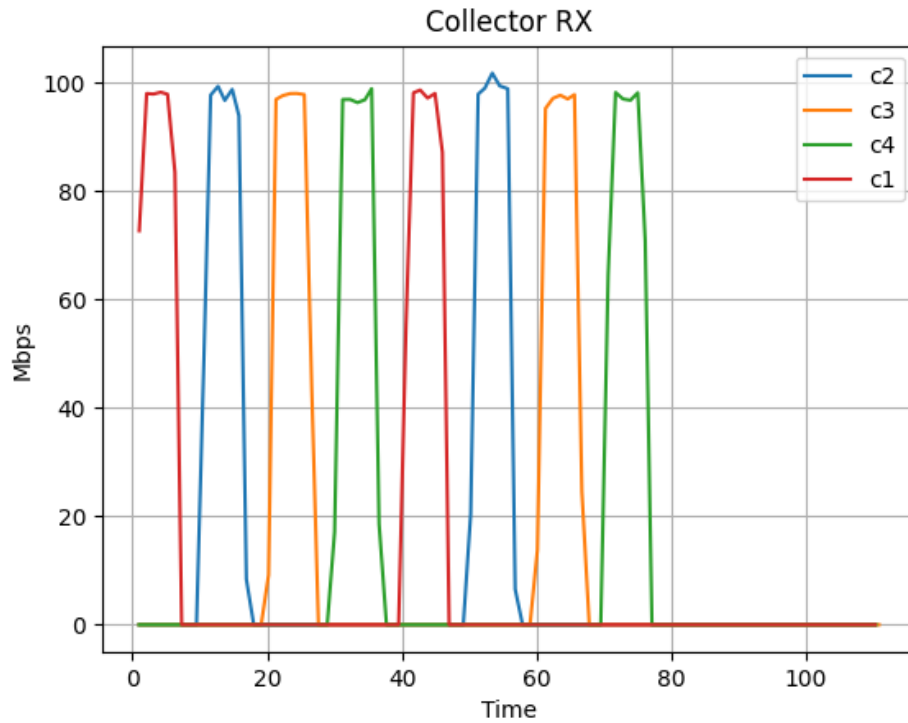


Figure 86: Scenario 4 – Simple strategy, Global result.

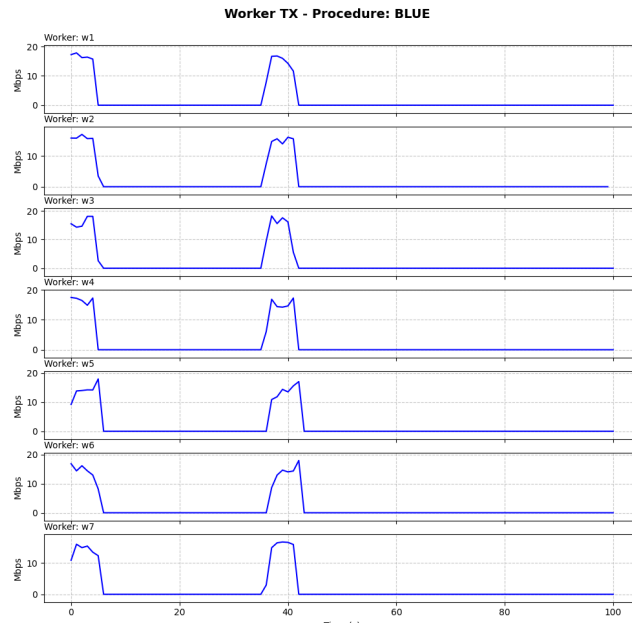


Figure 87: Scenario 4 – Simple strategy, Blue result.

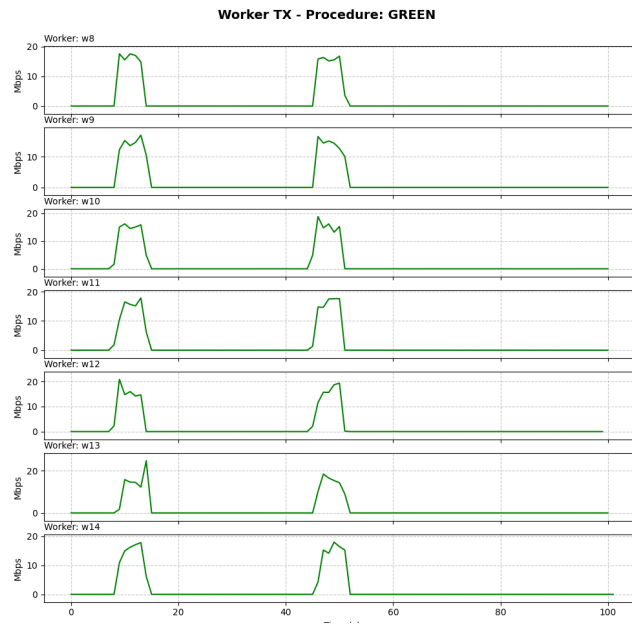


Figure 88: Scenario 4 – Simple strategy, Green result.

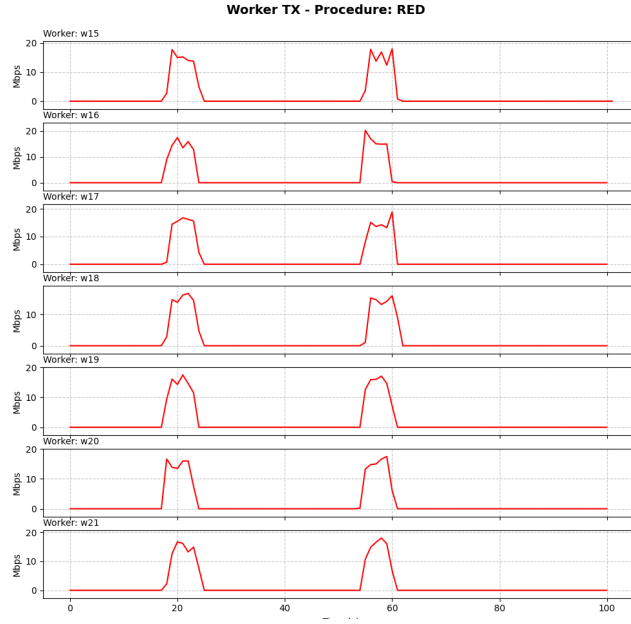


Figure 89: Scenario 4 – Simple strategy, Red result.

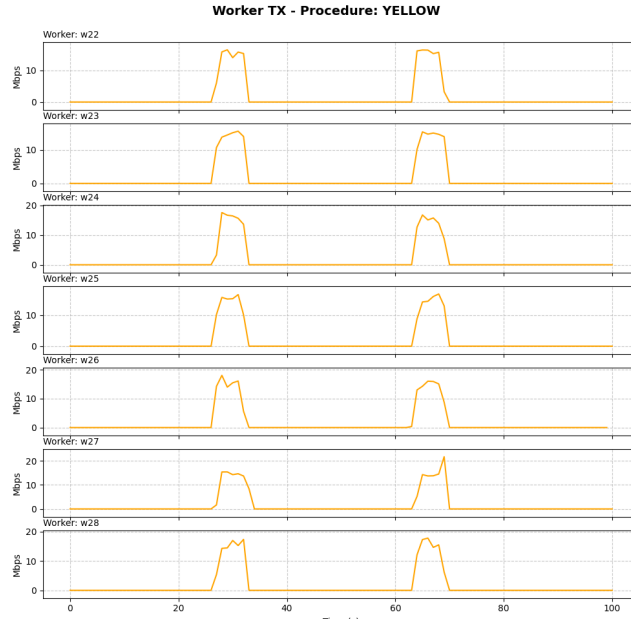


Figure 90: Scenario 4 – Simple strategy, Yellow result.

5.2 Normal Strategy

The following plots report all available outputs for the **normal** strategy in **scenario_4**.

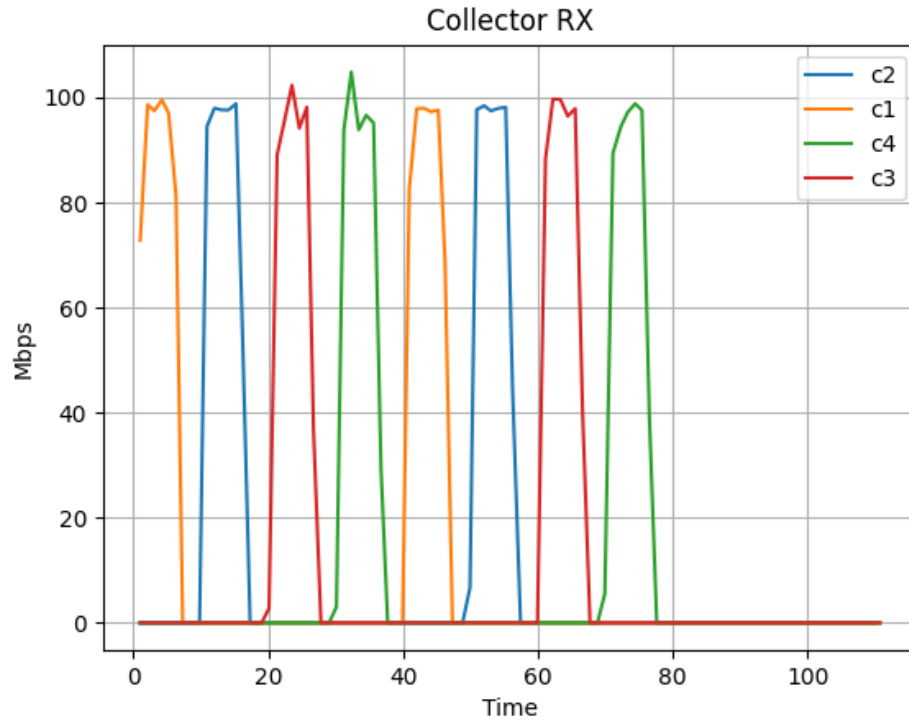


Figure 91: Scenario 4 – Normal strategy, Global result.

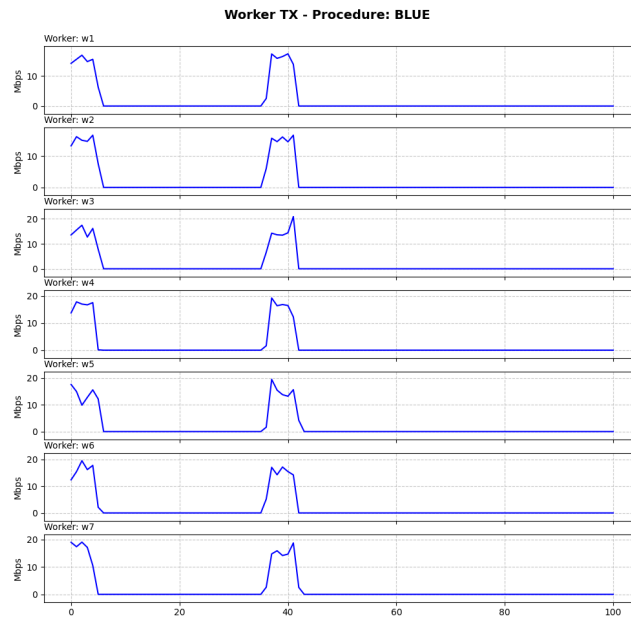


Figure 92: Scenario 4 – Normal strategy, Blue result.

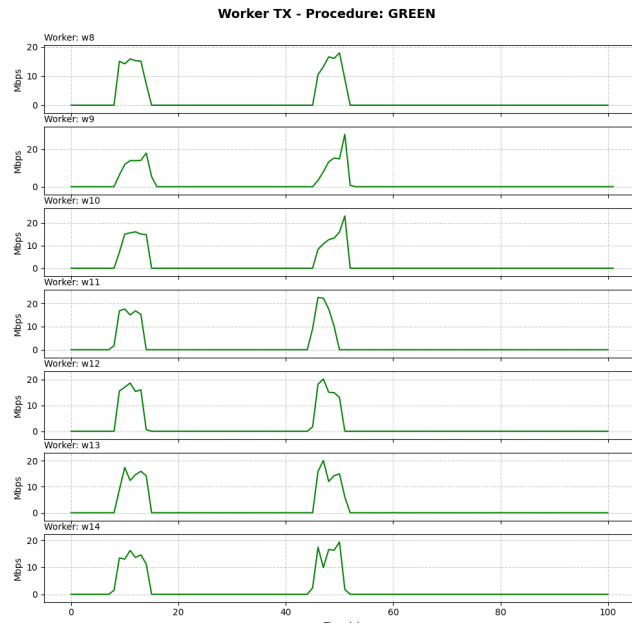


Figure 93: Scenario 4 – Normal strategy, Green result.

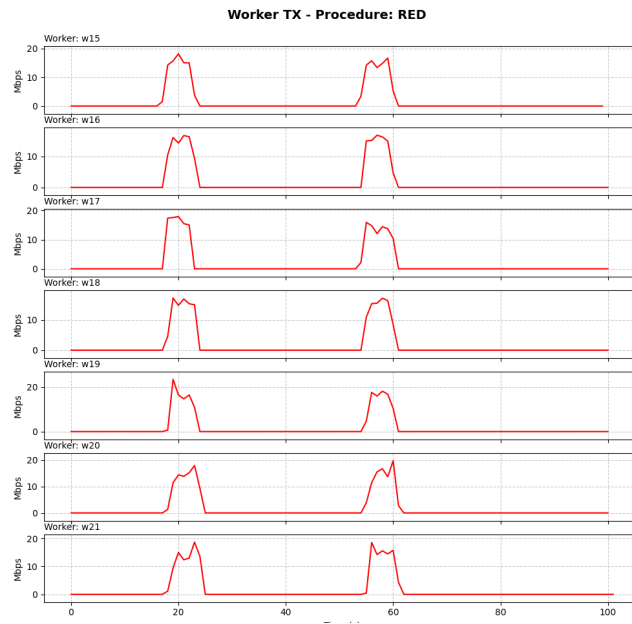


Figure 94: Scenario 4 – Normal strategy, Red result.

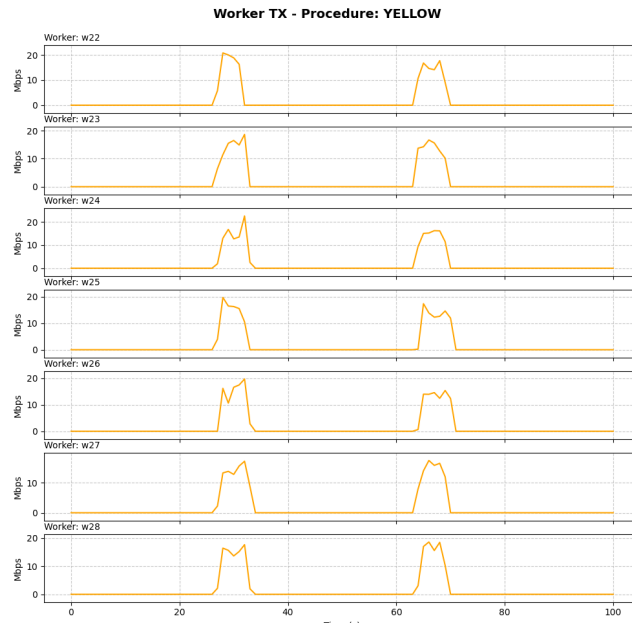


Figure 95: Scenario 4 – Normal strategy, Yellow result.

5.3 Advanced Strategy

The following plots report all available outputs for the **advanced** strategy in **scenario_4**.

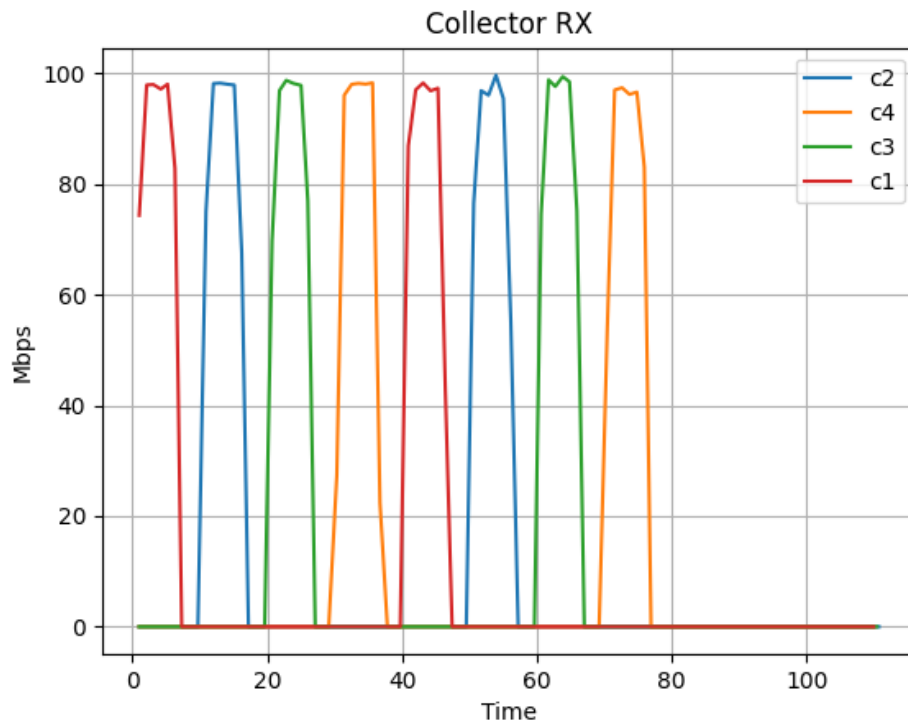


Figure 96: Scenario 4 – Advanced strategy, Global result.

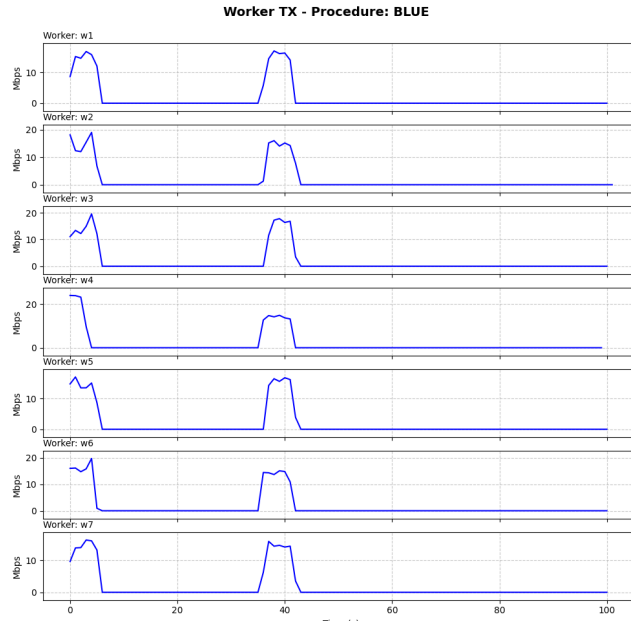


Figure 97: Scenario 4 – Advanced strategy, Blue result.

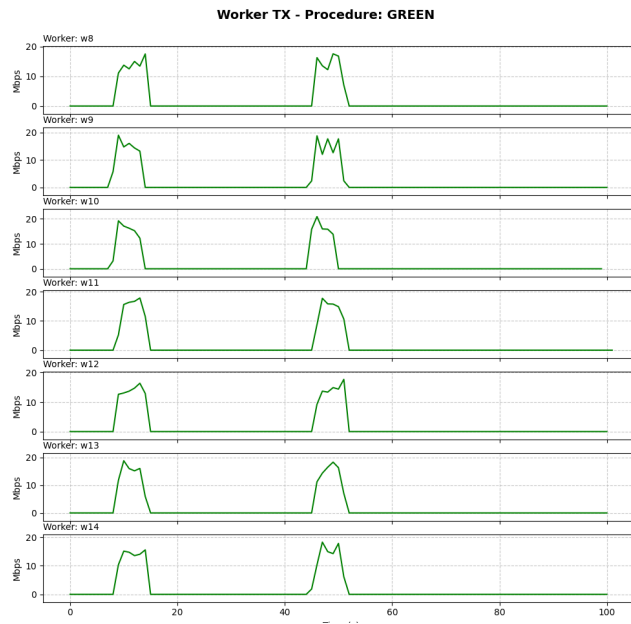


Figure 98: Scenario 4 – Advanced strategy, Green result.

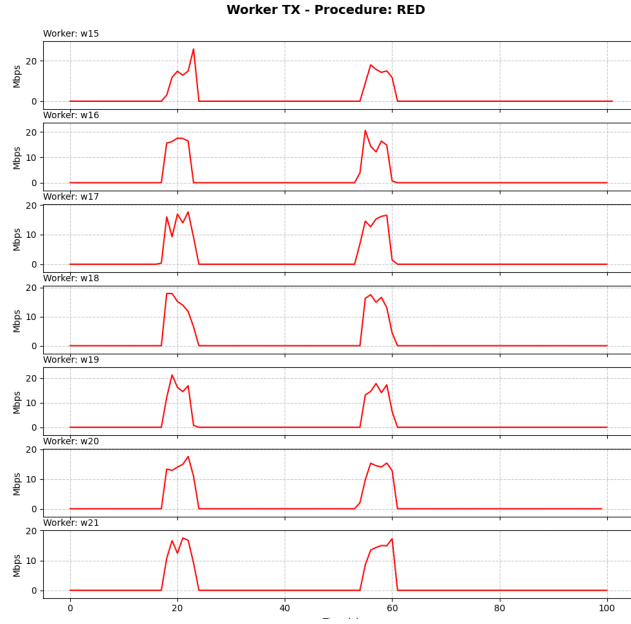


Figure 99: Scenario 4 – Advanced strategy, Red result.

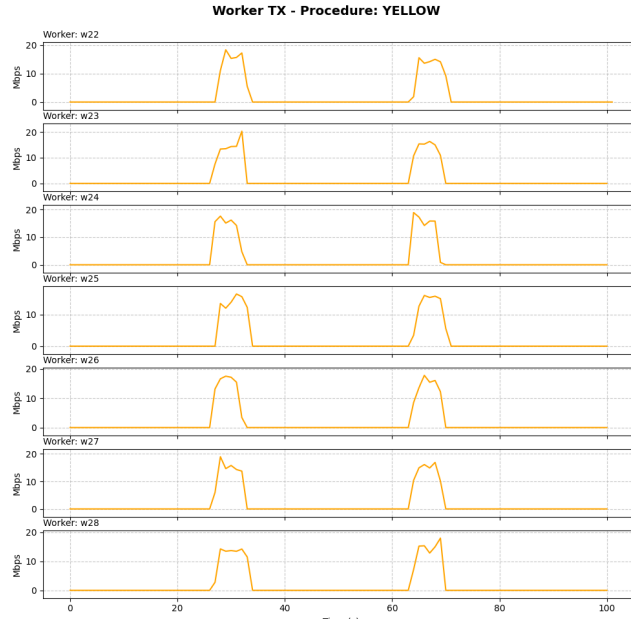


Figure 100: Scenario 4 – Advanced strategy, Yellow result.

5.4 Temporal Strategy

The following plots report all available outputs for the `temporal` strategy in `scenario_4`.

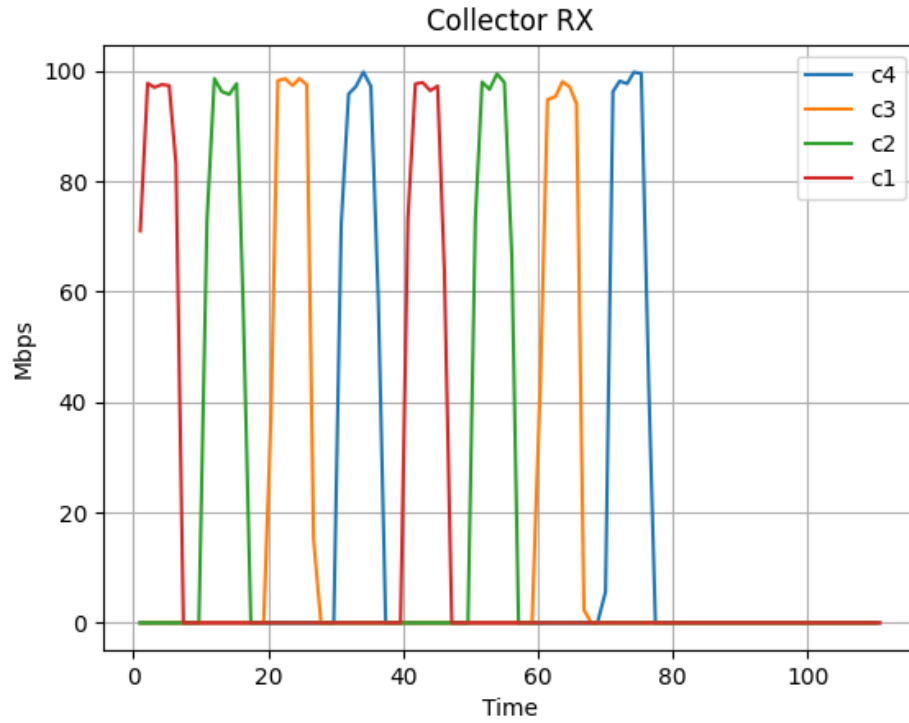


Figure 101: Scenario 4 – Temporal strategy, Global result.

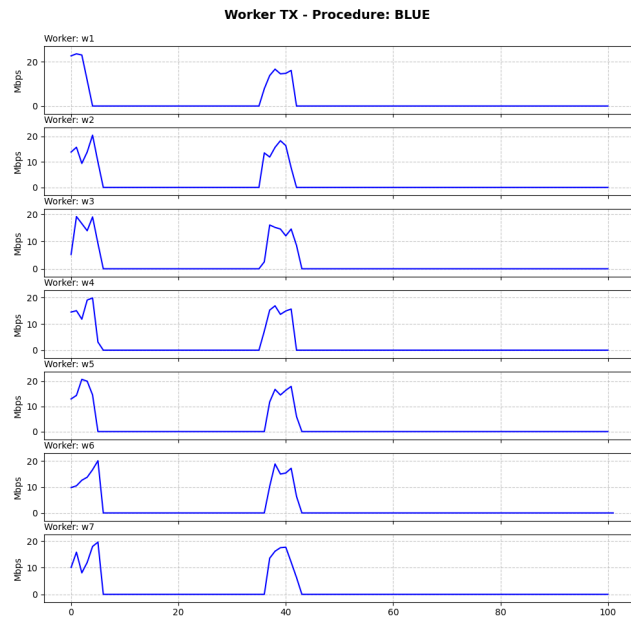


Figure 102: Scenario 4 – Temporal strategy, Blue result.

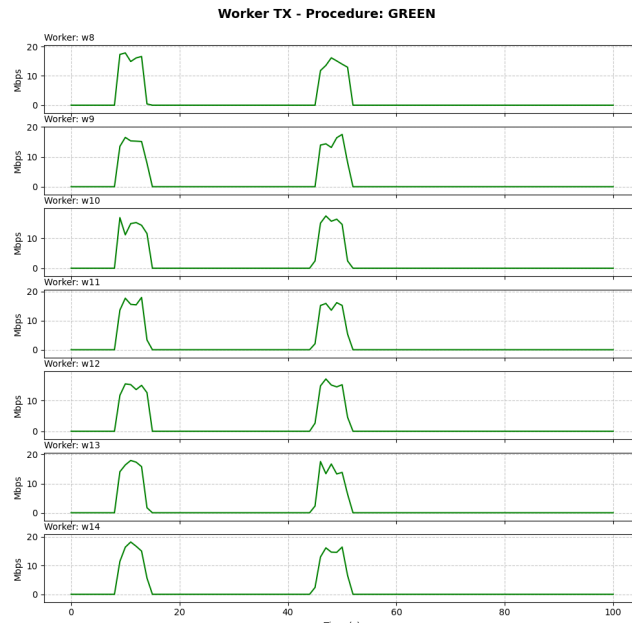


Figure 103: Scenario 4 – Temporal strategy, Green result.

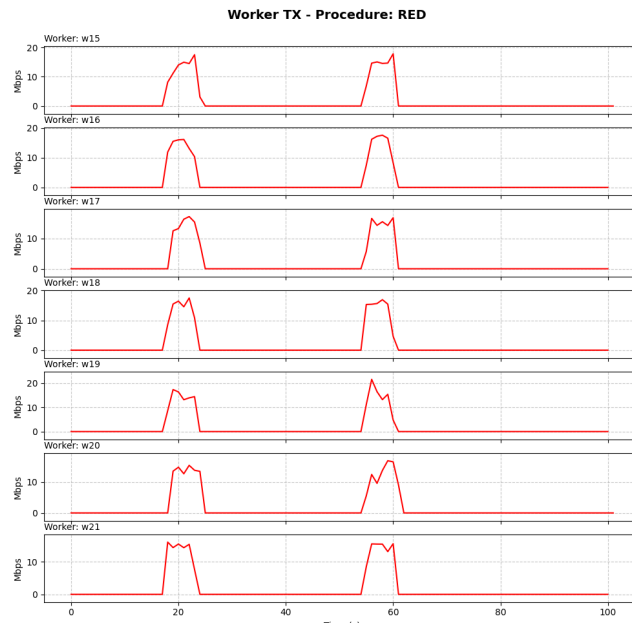


Figure 104: Scenario 4 – Temporal strategy, Red result.

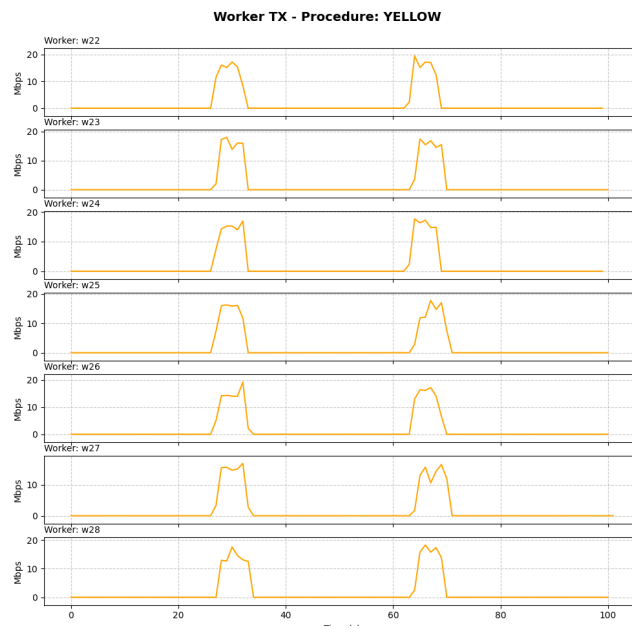


Figure 105: Scenario 4 – Temporal strategy, Yellow result.